Supplemental Information for:

Ancillary Consequences of Policy Interventions in the Presence of Disease-Based Poverty Traps

Current version: October 30, 2026

Contents

Results

Table S1: Percentage Changes from Baseline for Six Outcomes Across Policy Interventions 2
Figure S1: Percent changes in household income and infection prevalence, RoW Scenario 3
Figure S2: Synergistic and Trade-off Effects For Household Incomes, RoW Scenario 4
Figure S3: Synergistic and Trade-off Effects For Household Infection Prevalence, RoW Scenario . . 5
Figure S4: Percent changes in household income and infection prevalence, Loc Scenario 6
Figure S5: Synergistic and Trade-off Effects For Household Incomes, Local Scenario 7
Figure S6: Synergistic and Trade-off Effects For Household Infection Prevalence, Local Scenario . . 8
Figure S7: Synergistic and trade-off effects of concurrent policy interventions, Local Scenario . . . 9

Additional Details on Methods

Table S2:Parameterisation Summary: Economic Domain 10
Table S3:Parameterisation Summary: Public Health Domain 11
Table S4:Parameterisation Summary: Fisheries Domain 11
Estimations for the Local Economy 14
Estimating Parameter Values in the Epi Component 17
Estimating Parameter Values in the Ecological Component 20
Constructing the EEL Model 21
Solving the EEL Model 24

Appendices

A1 Assigning Slope and Median Parameter Values for β and γ 26
A2 Constructing the Next-Generation Matrix (NGM) Used to Derive $\mathcal{R}_0$ 27
A3 Equations of the EEL Model 31
A4 Parameterizing the MDA Intervention 34
A5 Robustness Check: Low and High Values of α 35
A6 Robustness Check: Alternative Specifications of Labour Supply Elasticity 39
A7 Robustness Check: Lagged Response of Labour Effectiveness to Infection 42
A8 Robustness Check: Net Migration in Response to Household Income 45
A9 Environmental Impact of Oil Palm 50
A10 Figure: Lifecycle of the *Schistosoma* parasite 51

Results

Table S1: Percentage Changes from Baseline for Six Outcomes Across All Policy Interventions

Outcomes	Year	Trade	MDA	FMP	TFP	MDA FMP	MDA TFP	FMP TFP	MDA FMP TFP
R0	1	Local	-22.78	0.79	-0.38	-22.28	-23.01	0.39	-22.52
		RoW	-22.78	-2.88	-0.46	-25.43	-23.08	-3.32	-25.71
	10	Local	-47.64	4.55	-4.56	-45.36	-49.06	0.29	-46.59
		RoW	-47.64	6.63	-5.00	-44.20	-49.34	1.22	-46.06
Infection Prevalence	1	Local	-19.59	0.54	-0.27	-19.07	-19.83	0.26	-19.32
		RoW	-19.59	-2.23	-0.32	-22.80	-19.90	-2.55	-23.11
	10	Local	-56.41	2.59	-3.17	-52.16	-59.33	-0.35	-54.58
		RoW	-56.41	3.50	-3.55	-50.13	-59.91	0.09	-53.66
Labour Time	1	Local	0.00	-0.07	0.43	-0.07	0.43	0.28	0.28
		RoW	0.00	-24.65	-0.06	-24.65	-0.06	-24.70	-24.70
	10	Local	-4.93	17.47	2.73	11.75	-2.32	21.93	15.99
		RoW	-4.94	27.21	-0.51	21.06	-5.41	26.57	20.43
Aggregate Output	1	Local	0.00	-1.53	0.94	-1.53	0.94	-0.60	-0.60
		RoW	0.00	-5.97	0.85	-5.97	0.85	-5.12	-5.12
	10	Local	0.03	0.69	10.03	0.72	10.06	10.79	10.83
		RoW	0.02	2.34	9.30	2.36	9.33	11.59	11.62
Fish Stock	1	Local	0.00	12.24	-0.11	12.24	-0.11	12.17	12.17
		RoW	0.00	17.67	0.01	17.67	0.01	17.68	17.68
	10	Local	-0.05	48.51	-3.23	48.47	-3.29	45.86	45.82
		RoW	-0.04	42.57	0.26	42.54	0.22	42.73	42.69
Harvest	1	Local	0.00	-17.24	0.15	-17.24	0.15	-17.14	-17.14
		RoW	0.00	-24.88	-0.02	-24.88	-0.02	-24.90	-24.90
	10	Local	-0.02	12.45	-0.83	12.45	-0.85	12.79	12.79
		RoW	-0.02	12.63	0.07	12.62	0.06	12.61	12.61

Notes: In the *Local* trade scenario, fishing sector output prices are endogenously determined. In the *RoW* trade scenario, fishing sector output prices are exogenous to changes in the local economy.

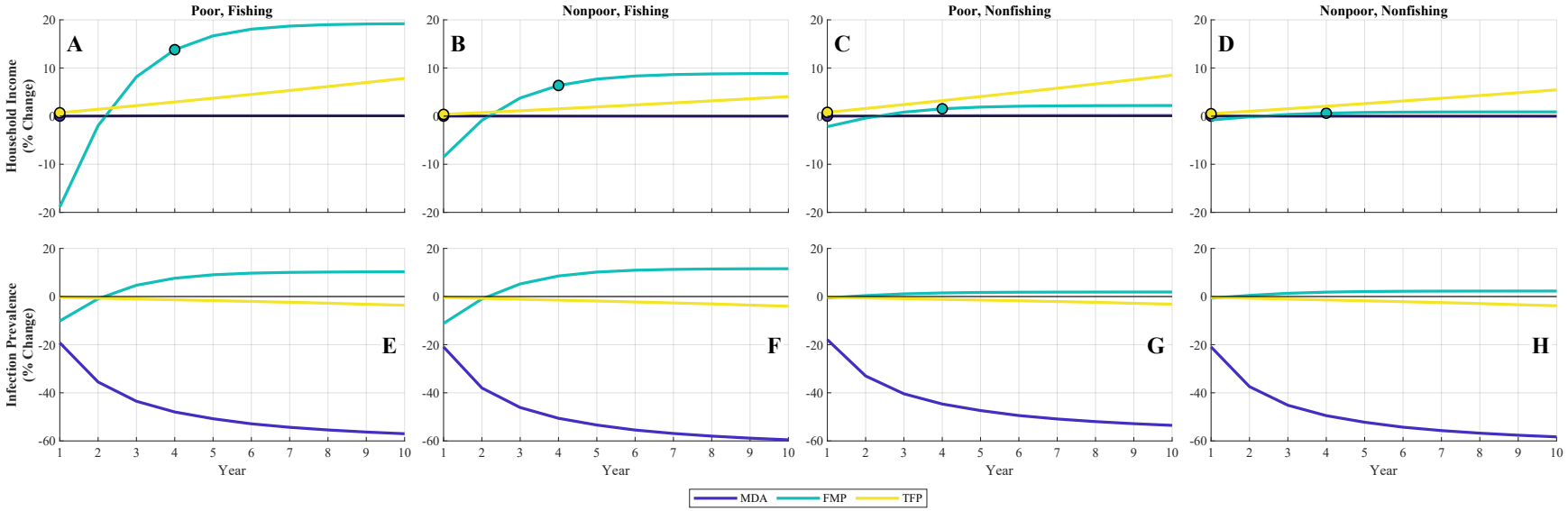


Figure S1: Percent changes in household income and infection prevalence relative to baseline for single-domain policy interventions. Fishing sector output prices are *exogenous* to changes in the local economy. Line markers on income figures indicate year in which NPV income turns positive.

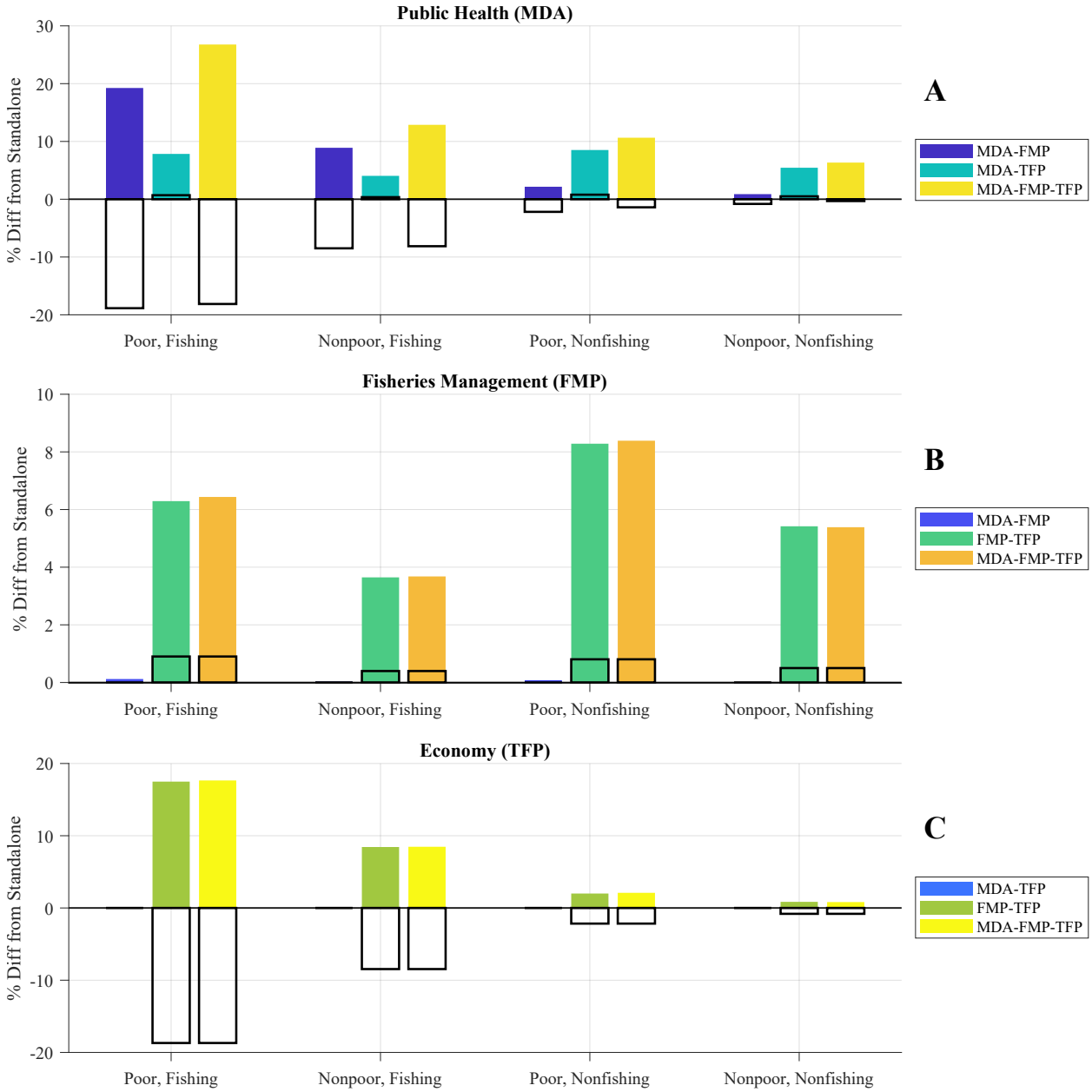


Figure S2: Synergistic and trade-off effects of concurrent policy interventions for household incomes. Fishing sector output prices are *exogenous* to changes in the local economy. Each panel represents the percentage change in household income when additional policies are added to a single-domain policy intervention. Results for Year 1 are shown as bars with black outlines and no colour fill. Results for Year 10 are shown as coloured bars with no outline. Each row represents a different domain policy perspective: MDA, FMP, and TFP.

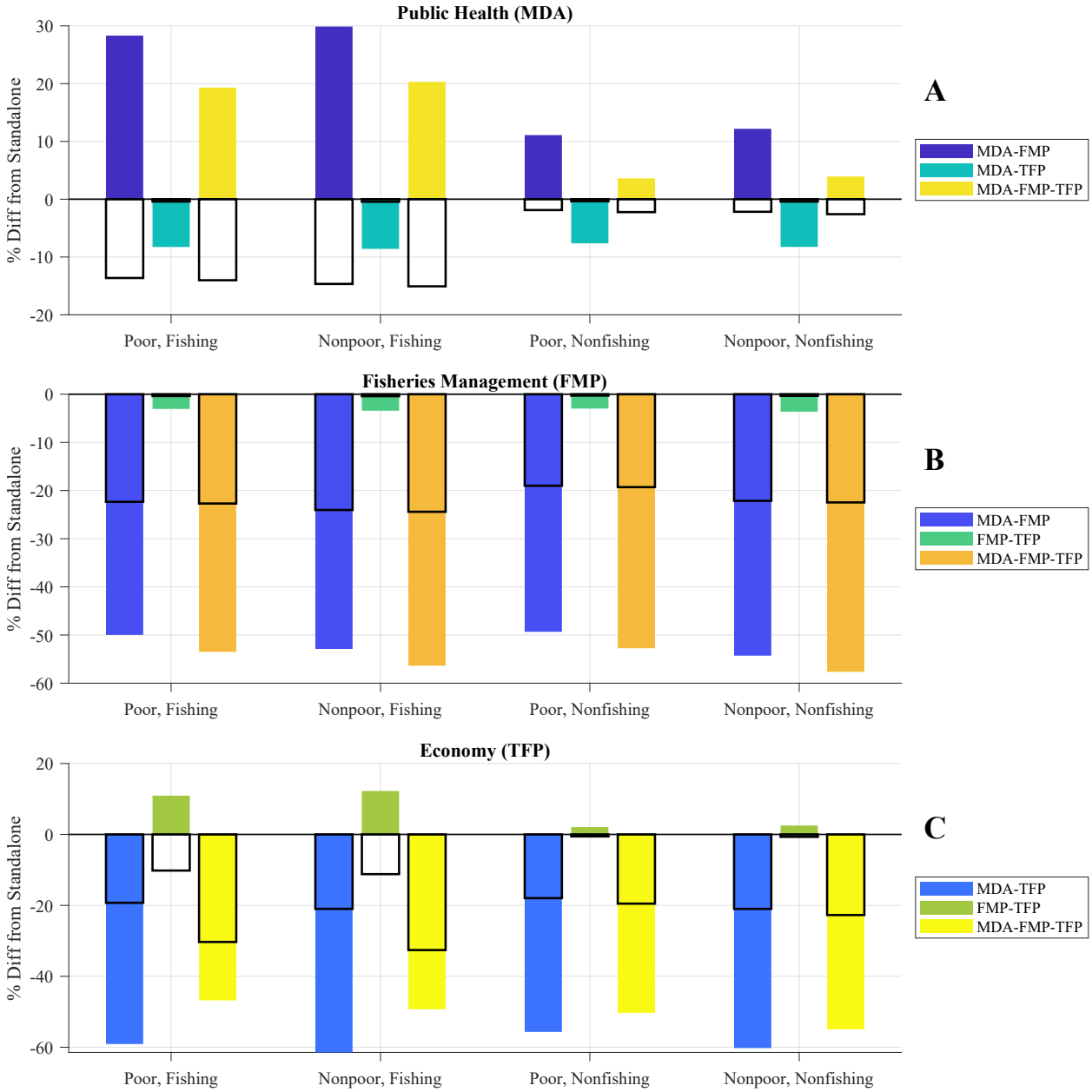


Figure S3: Synergistic and trade-off effects of concurrent policy interventions for household infection prevalence. Fishing sector output prices are *exogenous* to changes in the local economy. Each panel represents the percentage change in household infection prevalence when additional policies are added to a single-domain policy intervention. Results for Year 1 are shown as bars with black outlines and no colour fill. Results for Year 10 are shown as coloured bars with no outline. Each row represents a different domain policy perspective: MDA, FMP, and TFP.

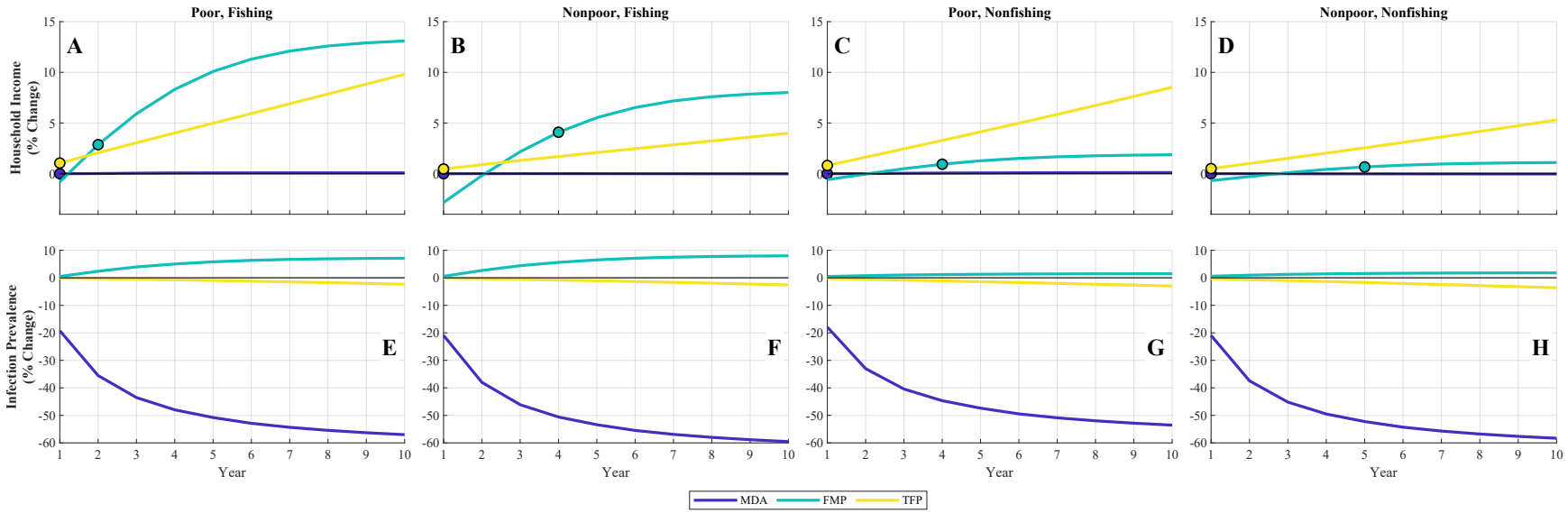


Figure S4: Percent changes in household income and infection prevalence relative to baseline for single-domain policy interventions. Fishing sector output prices are *endogenously* determined within the local economy. Line markers on income figures indicate year in which NPV income turns positive.

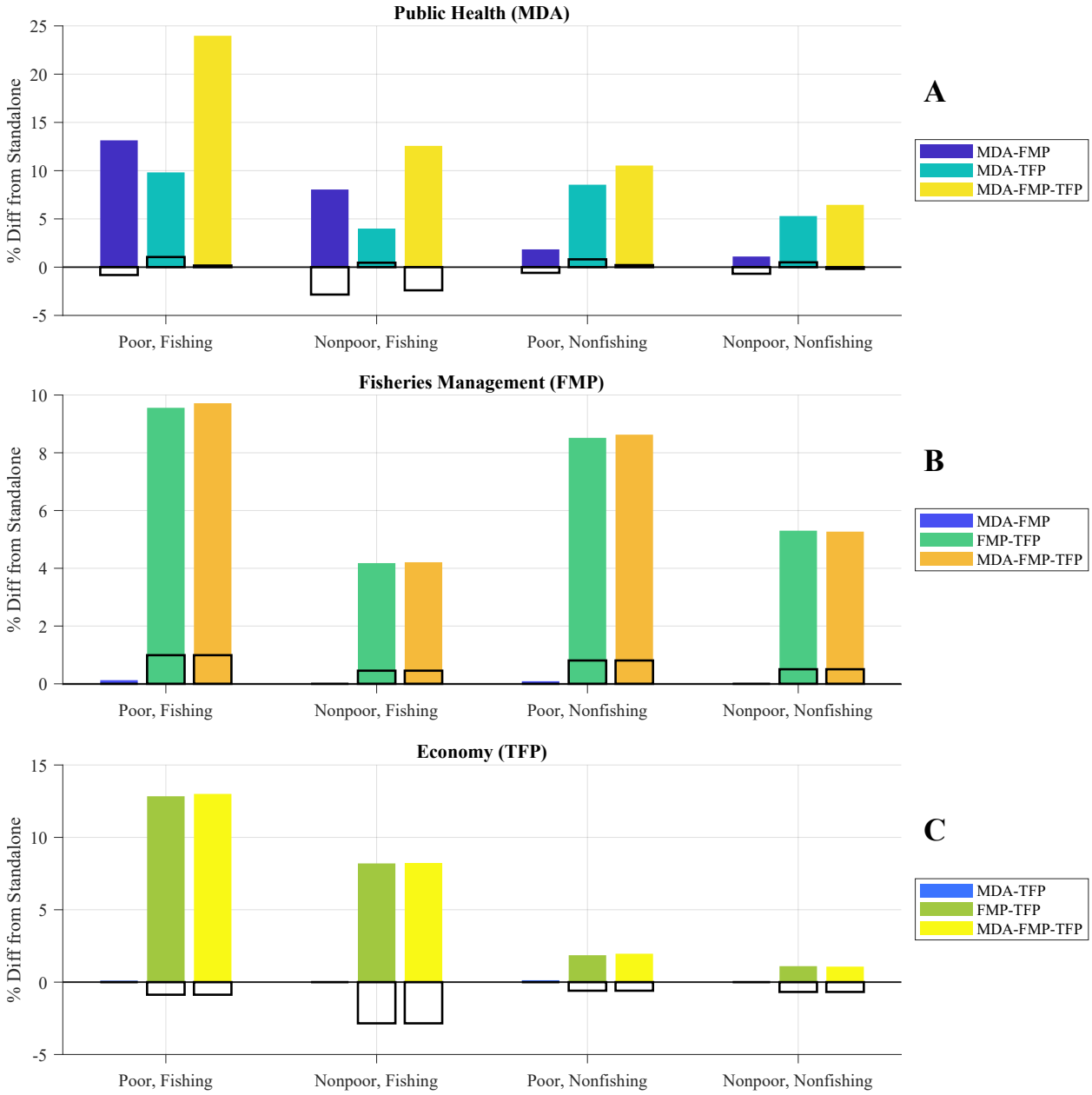


Figure S5: Synergistic and trade-off effects of concurrent policy interventions for household incomes. Fishing sector output prices are *endogenously* determined within the local economy. Each panel represents the percentage change in household income when additional policies are added to a single-domain policy intervention. Results for Year 1 are shown as bars with black outlines and no colour fill. Results for Year 10 are shown as coloured bars with no outline. Each row represents a different domain policy perspective: MDA, FMP, and TFP.

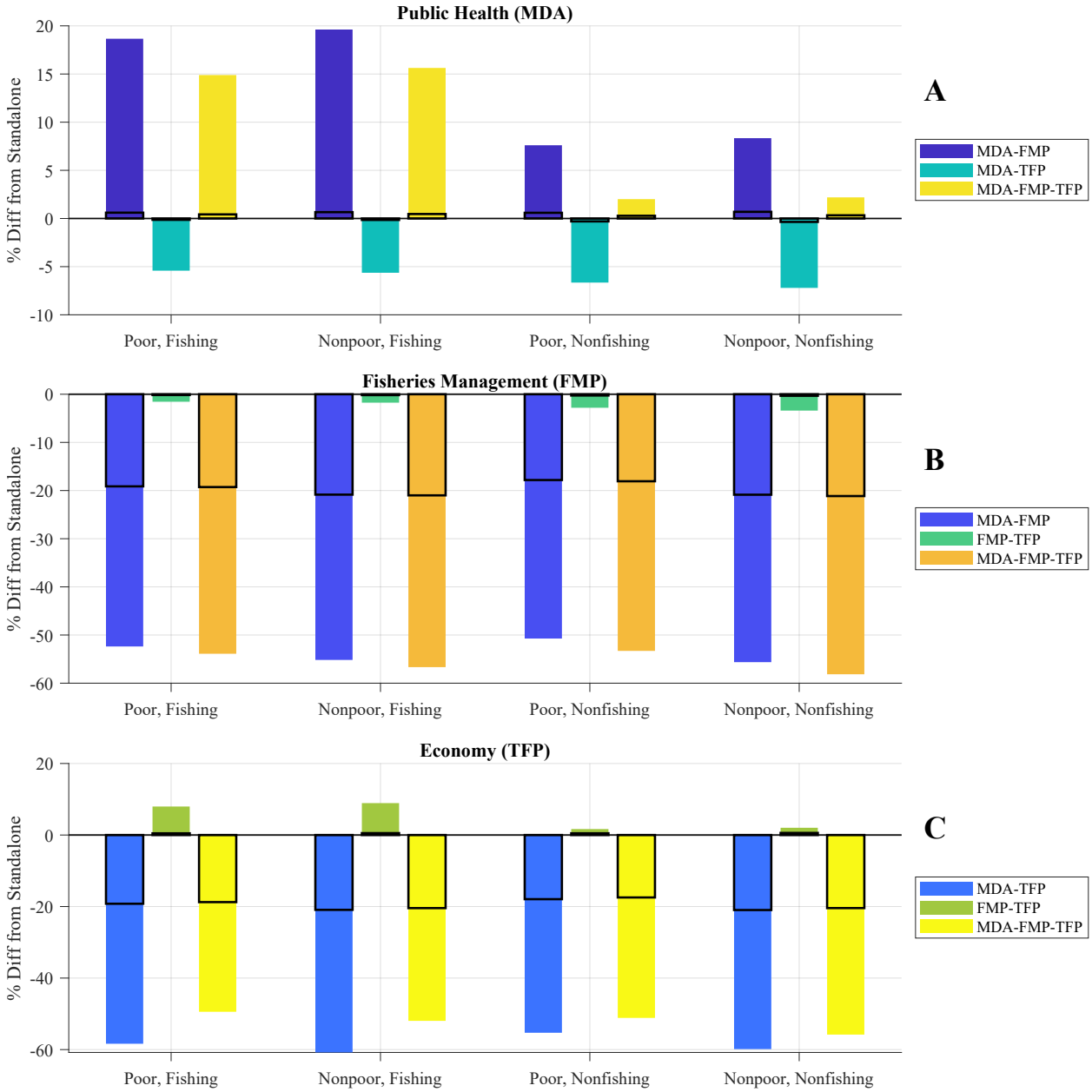


Figure S6: Synergistic and trade-off effects of concurrent policy interventions for household infection prevalence. Fishing sector output prices are *endogenously* determined within the local economy. Each panel represents the percentage change in household infection prevalence when additional policies are added to a single-domain policy intervention. Results for Year 1 are shown as bars with black outlines and no colour fill. Results for Year 10 are shown as coloured bars with no outline. Each row represents a different domain policy perspective: MDA, FMP, and TFP.

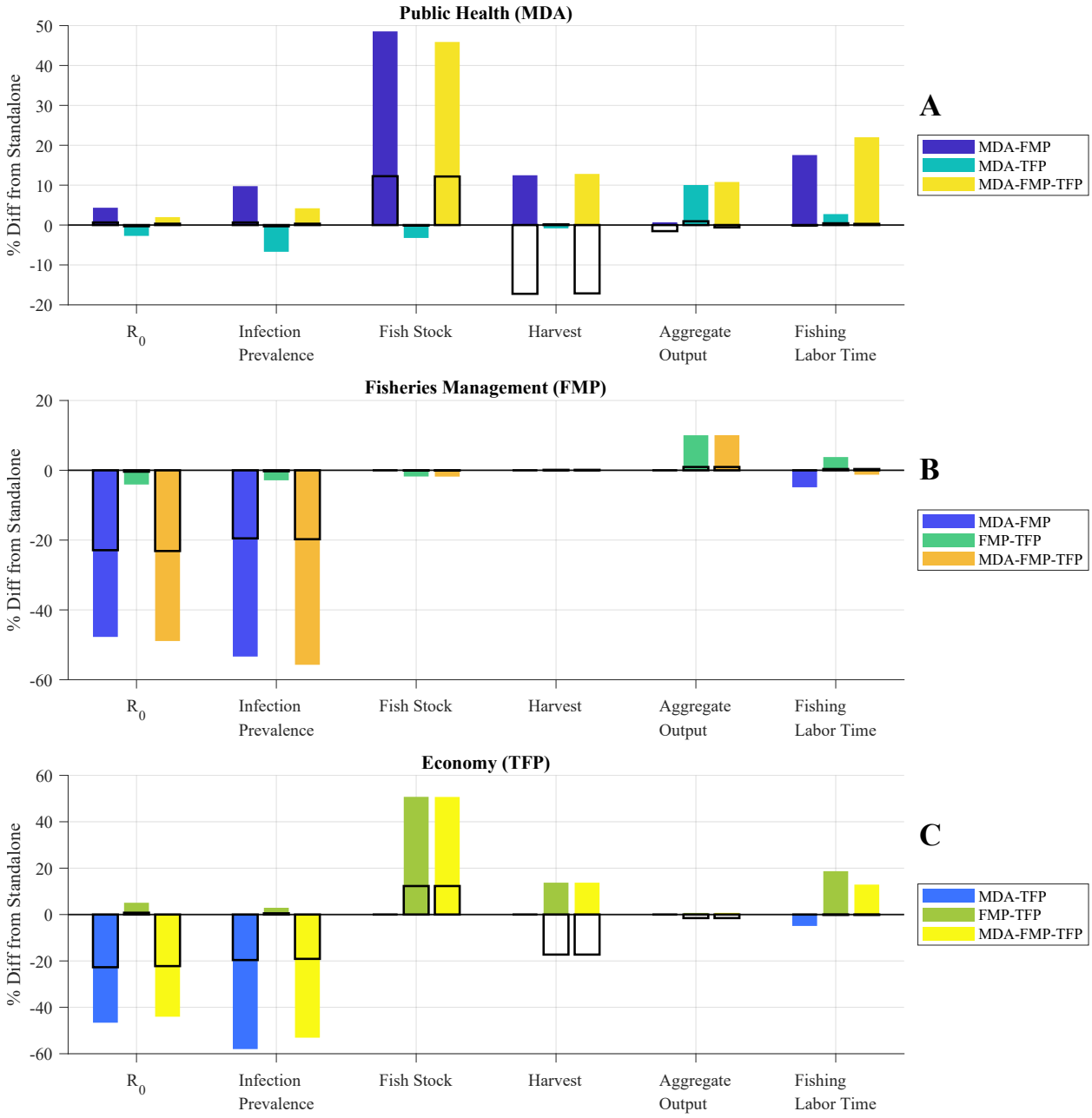


Figure S7: Synergistic and trade-off effects of concurrent policy interventions when fishing sector output prices are *endogenously* determined within the local economy. Each panel represents the percentage change in outcomes when additional policies are added to a single-domain policy intervention. Results for Year 1 are shown as bars with black outlines and no colour fill. Results for Year 10 are shown as coloured bars with no outline. Each row represents a different domain policy perspective: MDA, FMP, and TFP.

Additional Details on Methods

Parameterization Summary

Table S2: Summary of Economic Domain Parameters

Parameter	Description	Source / Calibration Procedure	SI Ref.
$\delta_{j,s}$	Production function output elasticities (labour, capital, land, inputs) by sector	Econometrically estimated from survey data using logged Cobb-Douglas specifications (inverse hyperbolic sine transform for zero values)	Table S4
Expenditure shares	Household budget shares by consumption category and household type	Seemingly unrelated regression (Stata v14.0) on survey data	Table S5
$popsh(h)$, $sav(h)$, $trout(h)$, $expout(h)$	Population shares, savings, net transfers, and outside expenditure shares by household	Calculated directly from survey data	Tables S8, S10
Labour supply elasticity	Responsiveness of sectoral labour supply to wages (assumed highly elastic)	Local unemployment/underemployment evidence and prior studies [1, 2]	p. 16
Substitution / trade elasticities	Elasticity of substitution between consumption goods (3); trade elasticity for fish (8)	Trade elasticity reflects shared regional sourcing of imported and local fish; both values govern the strength of the local price-feedback mechanism (see SI text)	Table S9

Notes: Page references point to the location in the SI where the parameter's value or derivation is discussed in the text; table references point to the SI table containing the underlying estimates.

Table S3: Summary of Public Health Domain Parameters

Parameter	Description	Source / Calibration Procedure	SI Ref.
α	Reduction in labour effectiveness due to infection (0.15, preferred)	Literature range; robustness checked for low/high values	p. 35
β_{min}, β_{max}, γ_{min}, γ_{max}	Bounds on the exposure rate and parasite mortality rate	Literature values [3, 4]	p. 18
μ	Snail-host parasite mortality rate (1.7×10^{-2})	Literature value [3, 4]	p. 18
χ	Human-to-snail transmission rate, assumed equal to β	Modelling assumption (shared exposure/contamination behaviour) [3, 5]	p. 18
Z_{med}, Z_{slope}	Sigmoid midpoint/slope for $\beta(Z)$, $\gamma(Z)$	World Bank per-capita GDP distribution (2017), scaled to the local economy; slope set following prior work [6, 4]	p. 26
ϵ_h	Household exposure-contamination risk parameter	Four survey measures (water collection time, water source, toilet facility, inverse per capita expenditure), normalised and averaged	Tables S6, S7
τ_h	Household exposure time	Fixed component from survey data plus time-varying component from fishing labour time, normalised to baseline	Table S7

Notes: Page references point to the location in the SI where the parameter's value or derivation is discussed in the text; table references point to the SI table containing the underlying estimates.

Table S4: Summary of Fisheries Domain Parameters

Parameter	Description	Source / Calibration Procedure	SI Ref.
δ_{fstock}	Output elasticity of the fish stock in the fishing production function (0.645)	Literature value [7]	p. 15
r_{fstock}	Annual growth rate of the composite fish stock (1.06)	Literature value [8]	p. 20
K_{lake}	Carrying capacity of the local fishing area (20,139 metric tons)	Derived from the Ugandan portion of Lake Victoria's carrying capacity, scaled by the estimated local fishing area (GIS-based) [8]	p. 20

Notes: Page references point to the location in the SI where the parameter's value or derivation is discussed in the text; table references point to the SI table containing the underlying estimates.

Household Survey Data

The survey data used for this study was collected in 2017 by a team of researchers from the Ugandan Ministry of Agriculture (MAAIF), the International Fund for Agricultural Development, and the University of California-Davis. The survey was conducted on Bugala Island in Kalangala District, Uganda. Bugala Island is the largest and most populated island in the district. The district government and the vast majority of businesses in the district are located on Bugala island.

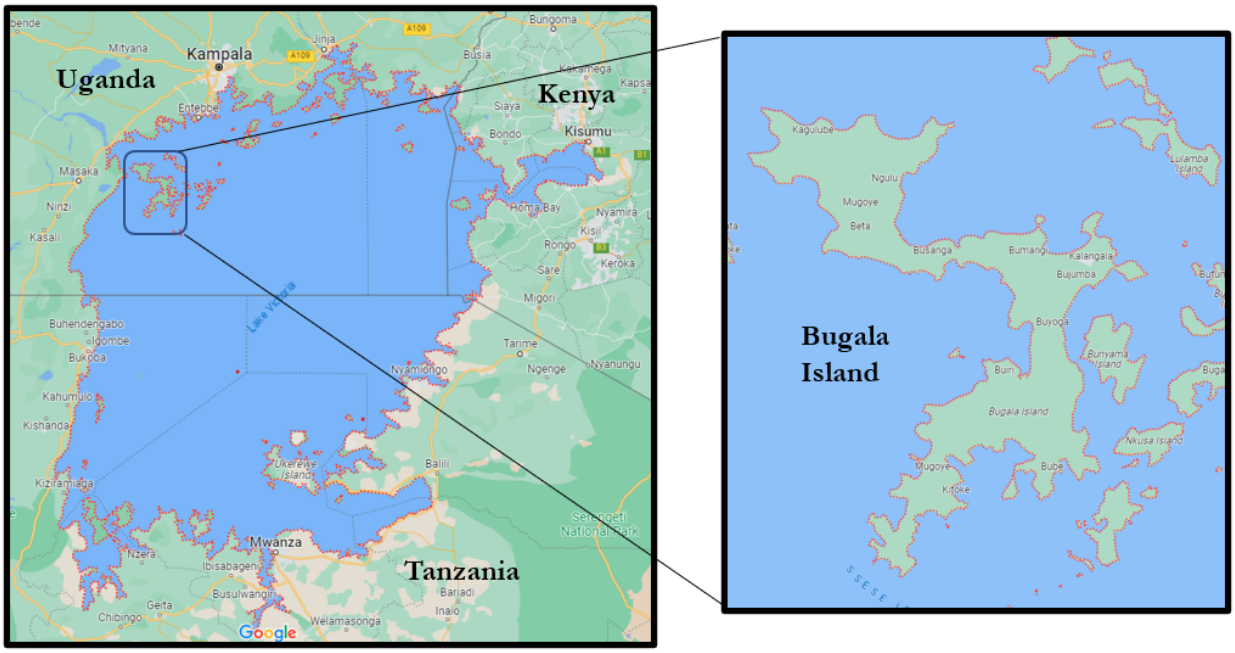


Figure S8: Map of Lake Victoria with bordering countries. Inset map showing location of Bugala Island in Lake Victoria. Produced using *QGIS ver.* 3.22.14.

A team of 14 local enumerators were trained over a period of 5 days, including two half-days of pilot-testing combined with training. A total of 511 households and 284 businesses in 39 villages were interviewed over a period of 3 weeks. Villages were randomly selected from a list of village names provided by the district government. Within each village, households were randomly selected from a list provided by village leaders. Without a master business list, enumerators were instructed to interview all businesses that consented and had not participated in the household survey. The business interview took approximately 15 minutes, and business owners were encouraged to pause the interview as needed so that business operations were not interrupted. Overall, less than 1% of households and businesses did not consent to interviews.

The survey questions were designed to gather detailed information on the household consumption and production activities of households and businesses on the island. The household survey consisted of several modules:

- An individual roster included questions related to education, health, employment and time-use questions.
- A household-level expenditure module contained questions about consumption and point of purchase for: purchases within the past two weeks (e.g., food items and services such as bars, restaurants, and transportation), purchases within the past month (e.g., utilities and fuel), and purchases within the past year for less frequent purchases (e.g., home maintenance and repair).
- Additional household-level modules included questions on dwelling characteristics, assets, and savings and remittances.

The household survey also included one module for each of the productive activities in the local economy: fishing, oil palm, other crops, livestock, and household-run businesses, along with a filter question asking

whether or not the household participated in the activity. Each production module included questions on output levels, input use, and other costs of production. To ensure data compatibility, the questions in the business survey were identical to the questions used in the business module in the household survey. Further details of the data collection effort can be found in Taylor *et al.*(2019) [9].

Table S5: Summary Statistics for the Four Household Types

Household (Observations)	Household Size	Age of Household Head (Years)	Dependency Ratio	Expenditures	Education, Household Head (Years)
Poor, Fishing	5.2	40.4	0.399	2,426	7.3
(n=14)	(2.4)	(10)	(0.254)	(856)	(4.0)
Nonpoor, Fishing	4.4	38	0.39	10,980	7.7
(n=70)	(2.5)	(11.1)	(0.231)	(6,173)	(4.2)
Poor, Nonfishing	5.1	42	0.495	2,225	6.7
(n=172)	(2.7)	(13.7)	(0.247)	(852)	(4.5)
Nonpoor, Nonfishing	3.3	37.9	0.304	9,399	8.6
(n=255)	(2.1)	(12.3)	(0.286)	(5,639)	(4.2)
Total	4.1	39.4	0.382	6,816	7.8
(n=511)	(2.5)	(12.7)	(0.278)	(5,791)	(4.4)

Notes: Averages for each measure are reported with standard deviations in parentheses. Daily per capita expenditures are reported in Ugandan Shillings.
Source: Original data.

Defining Representative Households

Using the survey data, we define four household types, or representative households, by classifying households in the survey data as either fishing or non-fishing, and as either poor or nonpoor. To classify fishing households, we rely on the initial screening question for the interview, which asks whether anyone in the household has engaged in fishing activities in the past 12 months.

To define poverty status, we use data from the household consumption module of the survey to calculate per capita consumption for each household. [1] We then define a daily per capita poverty line of $1.04, converted to a local currency amount of 3744 UGX using the exchange rate at the time of survey enumeration, and use this to assign poverty status to households with per capita consumption at or below this poverty line.[2] Table S5 shows summary statistics for sample households by classification.

Defining Productive Sectors

Information was collected for the six productive activities in the local economy: fishing, oil palm crops, food crops, livestock, and the service and retail business sectors. Oil palm crops are separated from other agriculture due to the relatively unique production process of this cash crop. Similarly, fishing is identified as a

[1] Ideally, we would define household poverty status using a measure of income for each household that could be constructed with the survey data. However, self-reported consumption data is widely regarded as more reliable than self-reported data on income, since the latter is more prone to errors resulting from either under- or over-reporting.

[2] This poverty line is in line with the national poverty line for Uganda, and well below the international poverty line of $1.90 per day established by the World Bank in 2015.[10]

separate activity due to its unique production process, which includes harvest of a renewable natural resource. Food crop production in the local economy consists of rice, *matoke*, cassava, maize, and other vegetables and fruits. Livestock production in the local economy consists mainly of cattle, poultry, goats, and pigs. Common businesses in the local service sector include bars and restaurants, barbers and hairdressers, and food processors. Grocery shops, corner shops and petty traders make up a majority of the retail businesses in the local economy. Subsistence farming is a common trait among households in the district, with approximately 65% of the labour force engaged in the activity as of the 2014 census [1].

Table S6: Household Participation by Sector

Household (Observations)	Oil Palm	Food Crops	Livestock	Fishing	Retail	Service
Poor, Fishing (n=14)	21.4%	50.0%	78.6%	100.0%	42.9%	28.6%
Nonpoor, Fishing (n=70)	24.3%	65.7%	74.3%	100.0%	15.7%	42.9%
Poor, Non-Fishing (n=172)	25.6%	59.9%	51.7%	0.0%	37.2%	23.8%
Nonpoor, Non-Fishing (n=255)	27.1%	47.5%	50.6%	0.0%	45.9%	36.9%

Source: Original data.

Estimations for the Local Economy

We specify functional forms for the equations in the LEWIE component and use the original household survey data to estimate them. For the parameters that we cannot estimate using the household data, we use values from the literature and, where applicable, undertake sensitivity analysis. In the text below, time-varying parameters have a t subscript. All other parameter values are fixed over the study period.

Production Functions

For the non-fishing sectors, we use a Cobb-Douglas production function to represent the relationship between the J final inputs used in the production process and final output. The production functions for the non-fishing sectors can be written as

$$QP_{s,t} = A_s \prod_{j=1}^{J} {f_{j,s,t}}^{\delta_{j,s}} \tag{S1}$$

where $QP_{s,t}$ is the total output produced in sector s in time t. A_s is a time-invariant total factor productivity (TFP) shift parameter. The parameter $f_{j,s,t}$ represents the amount of final input j used for production in sector s at time t. The exponent $\delta_{j,s}$ is the time-invariant output elasticity for final input j. We assume constant returns to scale in each of the non-fishing sectors. This means that $\delta_{j,s}$ is also the share of output attributable to final input j and that the sum of these output shares in each of the non-fishing sectors is

$$\sum_{j=1}^{J} \delta_{j,s} = 1 \ .$$

Production in the fishing sector is also modelled using a Cobb-Douglas production function. Accounting for the role that the fish stock plays in the fishing sector, we write the production function for this sector as

$$QP_{fish,t} = A_{fish} \prod_{j=1}^{J} f_{j,fish,t}^{\delta_{j,fish}} X_t^{\delta_{fstock}} \ . \tag{S2}$$

In contrast to the other productive sectors, we do not assume constant returns to scale in all inputs. We assume that returns to scale for all inputs *except the fish stock* are constant, and we assume a value of 0.645 for the output elasticity of the fish stock [7].

Intermediate Demands and Value-Added Output Price

Intermediate inputs are used in several of the productive sectors in the local economy. Examples include output from crop and livestock production that are used as inputs in production for the retail and service sectors. We model demand for intermediate goods using a Leontief production function [11]. This means that 1) there is no substitution among intermediate inputs and 2) the share of the value of total output attributable to intermediate inputs remains constant over time.

Using the value for intermediate inputs reported in the survey data, we calculate a share of the value added (VASH) by the final inputs to final output. We do this by subtracting the value of the intermediate inputs (INT) from the value of total output (output price (p_s) multiplied by quantity produced (QP_s)), and then divide this difference by the same value of total output:

$$VASH_s = \frac{p_s QP_s - INT_s}{p_s QP_s}$$

and, using the value of $VASH_s$, we calculate a value-added output price (pva) as

$$pva_s = p_s \times VASH_s$$

where the value-added price represents the value of the output produced using the final inputs *net of* the value of any intermediate inputs used in the production process. For the sectors that reported no use of intermediate inputs, $pva_s = p_s$. Even though output price and total output in a sector may vary over time in the model, we do not recalculate the value of $VASH_s$ in each time period and instead use baseline values of $VASH_s$ that remain fixed over the study period. Fuel is used in the fishing sector and is modelled as an intermediate input acquired from the service sector in the local economy. As with intermediate inputs in other sectors, we assume that $VASH_{fish}$ is constant over time.

Factor Demands

Using (S1), the profit-maximizing producer's quantity demanded for final inputs can be written as

$$f_{i,s,t} = \frac{pva_{s,t} QP_{s,t} \delta_{i,s}}{w_{i,t}} \tag{S3}$$

where the factor demands used in the local economy include effective labour, capital, land, and final inputs. Capital and land are classified as nontradeable inputs, which means that the supply of these inputs is fixed at the household level. Labour is tradeable between households in the local economy, resulting in changes in the supply of effective labour due to changes in the wage paid to labour. Our assumption that the local labour supply is highly elastic reflects the high unemployment in Kalangala District [1] and is consistent with previous studies in Africa [2]. Final inputs are tradeable outside the local economy, which means that the price for these inputs is fixed over the study period.

We estimate the production functions in each of the sectors using logged values for output and factor demands. To preserve observations that reported zero values for some (but not all) of the variables used in the regressions, we replace logged values using an inverse hyperbolic sine function. This method permits retention of data for

Table S7: Estimates for Production Function Output Elasticities

Variables	Stat	Fishing	Oil Palm	Crops	Livestock	Retail	Services
Effective Labour	coef.	0.33***	0.41***	0.14***	0.58***	0.14*	0.14***
	s.e.	(0.01)	(0.172)	(0.051)	(0.073)	(0.081)	(0.054)
Capital	coef.	0.67***	–	–	0.23***	0.17***	0.149*
	s.e.	(0.01)	–	–	(0.056)	(0.07)	(0.08)
Land	coef.	–	0.54***	0.77***	0.07***	–	–
	s.e.	–	(0.168)	0.059	(0.025)	–	–
Inputs	coef.	–	0.05*	0.1***	0.12**	0.68***	0.72***
	s.e.	–	(0.031)	(0.03)	(0.049)	(0.1)	(0.07)
Shift Param.	coef.	1.67	2.5	1.51***	7.89***	7.0***	8.0***
	s.e.	(1.172)	(1.901)	(0.35)	(0.742)	(1.08)	(0.89)
Observations		84	98	129	265	114	129

Notes: Robust standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1
Source: Original Data

households who were producers but did not use one of the factors of production, or households who engaged in production but reported zero output, possibly due to theft or loss of harvest.

The fishing sector only employs effective labour and capital, with capital accounting for two-thirds of the value-added production in that sector for the non-stock inputs. Producers in the oil palm and other crops sectors reported no use of machinery or other capital assets in the production process. The high share of capital in livestock production corresponds to the value of the animal stock used in the production process. Land value-added shares are larger for other crops than for oil palm, reflecting the relative importance of other inputs, including labour, in the production of oil palm. Inputs for both types of businesses contribute the majority of value-added to the production process.

Expenditure shares are estimated using a seemingly unrelated regression approach (*Stata* v14.0) which accounts for potential correlation in the error terms across equations. This means that for each household, a set of equations is estimated jointly, with one equation for each consumption category shown in Table S8. The equations are constructed with the category expenditure as the dependent variable and total expenditures as the independent variable. Household expenditures reflect both market transactions as well as the market value of goods produced and consumed within the household.

On average, we estimate that households allocate just under 10% of their budget on fish. Nonpoor fishing households are the exception, with a quarter of total expenditures going to fish consumption. Crops account for one-fifth of household consumption on average. Poor households consume a larger share of crops relative to nonpoor households, possibly reflecting the role of subsistence agriculture in household consumption. Livestock only accounts for 7% of household consumption. Retail and service businesses account for just under half of total consumption, with nonpoor-nonfishing households consuming above the average amount. Nonpoor households consume more outside purchases relative to poor households.

Table S8: Household Expenditure Share Estimates

Household	Fish	Crops	Livestock	Retail	Services	Outside
Poor, Fishing	0.08***	0.28***	0.07***	0.21***	0.23***	0.12**
(n=14)	(0.023)	(0.052)	(0.022)	(0.043)	(0.032)	(0.049)
Nonpoor, Fishing	0.25***	0.12***	0.05***	0.18***	0.22***	0.18***
(n=70)	(0.054)	(0.014)	(0.009)	(0.024)	(0.034)	(0.035)
Poor, Nonfishing	0.07***	0.27***	0.09***	0.21***	0.21***	0.16***
(n=172)	(0.009)	(0.016)	(0.009)	(0.014)	(0.012)	(0.018)
Nonpoor, Nonfishing	0.06***	0.17***	0.06***	0.24***	0.26***	0.21***
(n=255)	(0.006)	(0.009)	(0.004)	(0.012)	(0.012)	(0.015)
Total	0.08***	0.20***	0.07***	0.22***	0.24***	0.19***
(n=511)	0.007	0.008	0.004	0.008	0.008	0.011

Notes: Standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1
Source: Original Data

Estimating Parameters in the Epi Component

The parameters in the Epi component are assigned values following a process centred around identifying an interval for the basic reproductive number, R_0, that is grounded in previous research and is relevant for Schisto.

The expression for R_0 is

$$R_0 = \frac{\beta\chi \sum_{h=1}^{H} g_h \tau_h^E \tau_h^C \epsilon_h^E \epsilon_h^C}{\mu\gamma} \tag{S4}$$

with time subscripts suppressed. For the main analysis, we assume that χ is equal to and varies over time with the exposure rate parameter, β. This assumption is reasonable for diseases like Schisto, since it is likely that contamination occurs via the same behaviours that lead to exposure to the parasite [3, 5].

In order to establish an interval for R_0, we first assume that

$$\epsilon_h^E = \epsilon_h^C = 1 \qquad \text{and} \qquad \tau_h^E = \tau_h^C = 1$$

which states that each household type has the same amount of time and risk for both exposure to and contamination with the disease. The purpose of this step is to construct an appropriate interval for R_0 that is comparable to previous studies with populations that are assumed to be homogeneous with respect to exposure risk and time. Sokolow *et al.*(2015) [12] identifies parameter values based on an expected R_0 between 1 and 7, while Garchitorena *et al.*(2017) [4] assumes an endemic equilibrium R_0 of 3. Halstead *et al.*(2018) [13] identifies a maximum R_0 of 3.6 in their empirical study of the relationship between agrochemicals and densities of schistome-infected snails. As these studies implicitly assume that the population is homogeneous with respect the risk of exposure and contamination, it is appropriate to construct the interval for the R_0 in this study such that the maximum value for a homogeneous population is comparable to those used in previous studies.

The expression for R_0 simplifies to

$$R_0 = \frac{\beta^2}{\mu\gamma} \tag{S5}$$

after invoking the assumption that each household type has the same amount of exposure (and contamination) time and risk. Following previous studies [3, 4], we assign the value of $\mu = 1.7 \times 10^{-2}$. As discussed above, we model the parameters β and γ are both functions of aggregate output Z, which can vary over time. We establish minimum and maximum values for these parameters in line with values from previous studies, with $\beta_{min} = 5.8 \times 10^{-4}$ and $\beta_{max} = 5.8 \times 10^{-3}$ and $\gamma_{min} = 5.5 \times 10^{-4}$ and $\gamma_{max} = 5.5 \times 10^{-3}$ [3, 4]. We define values for the median and slope parameters for β and γ using country-level data from the World Bank (see page 26) [6].

We use the values of β_{max} and γ_{min} to calculate the maximum value of R_0. We choose these values to characterise that high rates of endemicity of the disease correspond to high rates of exposure to the disease and low rates of mortality of the parasite in the human population. Based on these parameter values, we calculate a maximum value of $R_0 = 3.6$. This is the same value found in Halstead *et al.*(2018) [13] and is in line with values from related studies noted above. It is worth emphasizing that this maximum value is based on a population with homogeneous risk of exposure and contamination. In the presence of *heterogeneous* risk of exposure and contamination across subpopulation groups, the observed R_0 may exceed this maximum value [3].

Exposure-Contamination Risk Parameters

The exposure and contamination risk parameters represent the risk of exposure to or contamination of the environment with the parasite that causes the disease that household h faces relative to other households. This risk may stem from differences in socioeconomic status, occupation, location, or ethnicity [14].

We assume that $\epsilon_h^C = \epsilon_h^E$ for each household and denote the Exposure-Contamination (E-C) risk parameter for each household as ϵ_h. This assumption states that for a given household, the risk of exposure to the disease is equal to the risk of contamination. As noted before, such an assumption is reasonable for diseases like Schisto, since it is likely that contamination occurs via the same behaviours that lead to exposure to the parasite [3, 5].

We identify values for ϵ_h using the survey data described above. Specifically, we combine data on household expenditures and data related to behaviours that are known to correlate with the disease: water collection and access to WASH facilities. Expenditure data is a proxy for *private* investment by the household in disease treatment and prevention. Data on water collection was collected in the individual roster of the survey and are aggregated at the household level. In the survey module on household assets, there are two questions related to access to clean WASH facilities. [3] For each source, Table S9 lists the question or section identifier found in the survey, the question content, and (if applicable) the recall window.

To calculate values of ϵ_h, we begin by producing raw averages at the household level for each of the four measures identified in Table S9. For each measure, a larger value correlates with an increased risk of exposure and contamination. Responses to the questions about main water source (WS) and access to toilet facilities (TF) were originally reported as single choices from a list of options. For these questions, we converted the data to a binary variable by reclassifying the choices as improved (0) or unimproved (1). We use values for water collection reported in the individual roster to create a measure of daily per capita water collection time for the household. We calculate an inverse measure of per capita household expenditures by dividing per capita household expenditure aggregate values into 1. Raw averages exclude outliers that are defined at the 97th percentile based on expenditures.

[3] Domestic production, which includes activities like water collection, cooking, cleaning, and child-rearing, is not identified as a productive activity in the LEWIE component for this study since, as in similar data collection efforts, domestic time use data were not gathered. As domestic production represents an additional transmission pathway for the disease in the local economy, future studies that adopt the modelling approach used in this study could begin to rectify this omission with a survey design that recognises these as value-adding activities for the local economy.

Table S9: Survey Data for Calculation of the E-C Parameter Values

Question Identifier	Question Content	Recall Window
TS001	Time spent collecting water	Hours in a typical day
HS003	Main water source (Household)	N/A
HS004	Access to Toilet Facilities	N/A
Household Expenditure Section		Various; Synchronized

Source: Original Data

Table S10: Summary Statistics and Estimated E-C measures

Household Type	Pop Share (%)	Water Collection	WS	TF	Expenditures (inverse)	ϵ_h
		WC	WS	TF	exp	
Poor, Fish	2.7	1.59	33.3%	16.7%	4.4×10^{-4}	1.064
		1.038	0.979	0.741	1.486	
Nonpoor, Fish	13.6	1.82	20.5%	20.5%	1.2×10^{-4}	0.876
		0.951	0.823	0.912	0.410	
Poor, Nonfish	33.7	2.35	36.8%	22.4%	4.5×10^{-4}	1.232
		1.122	1.081	0.996	1.536	
Nonpoor, Nonfish	49.9	1.63	33.0%	24.1%	1.4×10^{-4}	0.874
		0.851	0.971	1.072	0.477	

Notes: The E-C risk parameter for each household is shown in the rightmost column. For each household, raw averages for each measure are shown above the household's normalised value for that measure. Water Collection is measured in hours per day.
Source: Original Data.

For each of the four measures, we normalise the raw average for the household by dividing by the population-weighted mean. For example, the normalised value of water collection time (WC) for household h is calculated as

$$\epsilon_{WC,h} = \frac{WC_h}{\sum_{\tilde{h}=1}^{H} g_{\tilde{h}} WC_{\tilde{h}}}$$

where the summation term in the denominator is over all households. Dividing by the mean produces a unit-free value, which allows for aggregation across measures with different units (e.g., hours per day and expenditures). We then calculate ϵ_h as

$$\epsilon_h = \frac{1}{4}\Big[\epsilon_{WC,h} + \epsilon_{WS,h} + \epsilon_{TF,h} + \epsilon_{exp,h}\Big]$$

which, being a simple average, means that we assume that each measure is of equal importance in determining

the value of ϵ_h.

Overall, poor households reported spending more time than nonpoor households collecting water and using unimproved water supplies with greater frequency. Nonpoor fishing households reported a greater frequency of unimproved toilet facilities. Poor nonfishing households have the highest E-C risk parameter value at 1.232, which may reflect the impact of lower average expenditures for these households compared to poor fishing households. Both nonpoor household groups have approximately the same E-C risk parameter value, possibly reflecting tradeoffs between exposure to the disease via fishing and higher expenditures reported by fishing households.

Exposure Time

The measure of exposure time represents two of the primary ways in which individuals are exposed to the disease:household production activities like water collection and fishing labour time. Only fishing labour time varies over the study period. Household production activities correlated with exposure time are fixed.

To calculate the baseline levels of exposure time for each household, we calculate a fixed value, which consists of the values for water collection time identified in Table S10, and a variable value, which consists of fishing labour time supplied by the household, converted to daily values. Using the survey data, we estimate that fishing labour time accounts for approximately 80% of fishing household's baseline level of exposure time. The baseline value of τ_h is normalised to 1 for each household. We calculate exposure time for the household at time t as

$$\tau_{h,t} = \bar{\tau}_h + \frac{L_{h,fish,t}}{L_{h,fish,baseline}} \tag{S6}$$

where $\bar{\tau}_h$ is the fixed value of exposure time. The variable component of exposure time is calculated by dividing the amount of fishing labour time supplied by household h by the baseline amount of fishing labour time supplied by household h. The value of $L_{h,fish,t}$ is solved for using the amount of effective labour supplied by the household to the fishing sector and the household's infection status. For nonfishing households, exposure time remains constant over time. For fishing households, exposure time may change as a result of changes in their labour allocation across sectors.

Estimating Parameter Values in the Ecological Component

The state equation for the ecological component of the model is:

$$\dot{X} = X_t r_{fstock}(1 - \frac{X_t}{K_{lake}}) - HARV_t \, . \tag{S7}$$

The first parameter value from (S7) that we source from previous literature is r_{fstock}, the growth rate for the composite fish stock. We follow Kateregga *et al.*(2009) [8], which utilises a biological population model for the composite fish stock in Lake Victoria, by assuming $r_{fstock} = 1.06$.

The value for the carrying capacity of the local fishing area, $K_{lake} = 20,139$ metric tons, is based on the assumption that the stock of fish are essentially distributed uniformly across the lake and estimated as follows. We source an estimate of the carrying capacity for the Ugandan portion of Lake Victoria, 492,000 metric tons [8]. We estimate the share of the fishing area for the local economy by calculating the Ugandan portion (45%) of Lake Victoria total surface area to be $0.45 \times 68.800\,\text{km}^2 = 30.960\,\text{km}^2$. Next, we estimate a fishing area around Bugala Island of $1267\,\text{km}^2$ (see Figure S9).[4]



Figure S9: Fishing area around Bugala Island.

[4]The fishing area around Bugala Island was estimated by creating a polygon perimeter around the island with a distance of 10 km from the island shoreline, using *QGIS ver.* 3.22.14. The value of this distance was informed by conversations with local fisherman during the author's field visit in 2020.

We estimate the share of the Ugandan portion of the lake that is fished by the local economy to be

$$\frac{1267\,\mathrm{km}^2}{30.960\,\mathrm{km}^2} = 4.1\% \ .$$

We then multiply 4.1% by the estimate of 492,000 metric tons from Kateregga *et al.*(2009) [8], obtaining a value of 20,139 metric tons for the carrying capacity of the local fishing area. The baseline value of $HARV$ is assigned using the output from the baseline solution process of the EEL model, described next.

Constructing the EEL Model

There are a total of 360 equations that comprise the EEL model: 9 equations in the Epi component, 1 equation in the Eco component, and 350 equations in the LEWIE component, with a corresponding number of variables whose values are identified in the solution process (see page 31 for a complete list of model equations and corresponding variables). The objective of the process outlined in this section is to obtain baseline solutions for the EEL model such that equilibrium is reached within each component model and between component models. Absent any changes to parameter values in the EEL model, including those that result from a simulated policy intervention, the EEL model will remain in equilibrium. The EEL model is solved as a mixed complementarity problem (MCP) using the *MCP* solver in General Algebraic Modelling System (*GAMS*) software *ver.* 24.1.3.

Preparing the LEWIE Component for Analysis

Table S11: Exogenous Parameters Used to Solve for Baseline Solutions for the LEWIE Component.

Identifier	Definition
$fshare(f,g,h)$	Share of input factor f used in the production of output g by household h
$eshare(g,h)$	Share of household h's total expenditures allocated to consumption of good g
$sav(h)$	Savings by household h (share of total household income)
$trout(h)$	Net transfers out by household h (share of total household income)
$expout(h)$	Expenditures outside of the local economy by household h (share of total household income)

Notes: Relevant sets for each parameter, as are shown in parentheses in the table, are as follows: $f \rightarrow$ factor input; $g, gg \rightarrow$ produced good; $h \rightarrow$ household.

We assume a baseline value of 1 for output prices and factor prices, which allows us to interpret results as percentage changes in prices and wages relative to baseline [11]. We assume an elasticity of substitution between consumption goods of 3 for each household and a trade elasticity of 8 for the composite good, fish, which implies that imported fish are a close substitute for locally harvested fish. This assumption makes sense as imported fish most likely come from within the region and thus are likely coming from the same source, Lake Victoria, as locally harvested fish. We use the survey data to calculate population shares for each household type, which enter into the model as the parameter $popsh(h)$.

Endogenously Determined Parameter Values and Solutions to the Model

The remaining parameter values are endogenous to the LEWIE component of the model and are obtained during the process of solving for the baseline values of the EEL model, which are chosen to match the conditions of the baseline economy as observed in the data. Using the sampled values and model assumptions

Table S12: Model Assumptions for Solving for Baseline Solutions for the LEWIE Component

Identifier	Assumption
$p(g) = 1$	Output prices are equal to 1
$r = 1$ $w = 1$	Factor wages for capital (r) and labour (w) are equal to 1
$tr_elas(g) = 8$	Trade elasticity for good g is equal to 8; constant over time
$good_elas(h) = 3$	Consumption elasticity for household h is equal to 3; constant over time
$p_imp(g) = 1$	Import price of good g is equal to 1
$p_comp(g) = 1$	Composite of domestic and import prices of good g is equal to 1

Notes: Relevant sets for each parameter, as are shown in parentheses in the table, are as follows: $f \rightarrow$ factor input; $g, gg \rightarrow$ produced good; $h \rightarrow$ household; $dr \rightarrow$ draw.

described above, we calculate the values of the endogenous parameters and the solution values for the LEWIE component variables as identified in Table S13.

Table S13: Values, presented in order of calculation, that are either parameter values or initial guesses for obtaining baseline solutions for the LEWIE component.

Identifier	Description	Calculation
QC(g,h)	Monetary value of the quantity of good g consumed by household h	$[\mathbf{HHINC}(\cdot) - sav(\cdot) - trout(\cdot) - expout(\cdot)] \times \frac{eshare(\cdot)}{p(\cdot)}$
ID(gg,g,h)	Monetary value of the intermediate demand for good gg used in the production of final output g by household h	$\mathbf{QP}(\cdot) \times idsh(\cdot)$
FD(g,f,h)	Monetary value of the quantity demanded of input factor f in the production of good g by household h	$[\mathbf{QP}(\cdot) - \sum_{gg} \mathbf{ID}(\cdot)] \times fshare(\cdot)$
QVA(g,h)	Monetary value of quantity of good g produced by household h	$\sum_f \mathbf{FD}(\cdot)$
$tfpshift(g,h)$	Total Factor Productivity value for good g produced by household h	$\frac{\mathbf{QVA}(\cdot)}{\prod_f \mathbf{FD}(\cdot)^{fshare(\cdot)}}$
IMP(g)	Imports of good g	$\frac{1-domsh(\cdot)}{domsh(\cdot)} \times \sum_g \mathbf{QP}(g,\cdot)$
$vash(g,h)$	Final inputs' share of the total value of output g produced by household h	$\frac{\mathbf{QP}(\cdot) - \sum_{gg} \mathbf{ID}(\cdot)}{\mathbf{QP}(\cdot)}$
$endow(f,h)$	Household h's endowment of factor f	For Land and Capital : $\sum_g \mathbf{FD}(\cdot)$ For Labour : $popsh(\cdot) \times \sum_{g,h} \mathbf{FD}(\cdot)$

Continued on next page

Table S13 – continued from previous page

Calculated Values	Description	Calculation
$delta(g)$	Share parameter for the Armington function	$\frac{\mathbf{IMP}(\cdot)}{\sum_h \mathbf{QP}(\cdot)}^{tr_elas(\cdot)} \times \left[\frac{p_imp(\cdot)}{p(\cdot)} + \frac{\mathbf{IMP}(\cdot)}{\sum_h \mathbf{QP}(\cdot)}^{tr_elas(\cdot)}\right]^{-1}$
TQP_COMP(g)	Aggregate output of composite good g	$\mathbf{IMP}(\cdot) + \sum_h \mathbf{QP}(\cdot)$

Notes: Parameters are written in italic font. Variables are capitalized and written in bold font. Relevant sets for each parameter or variable are represented in parentheses as follows: $f \rightarrow$ factor input; $g, gg \rightarrow$ produced good; $h \rightarrow$ household; $dr \rightarrow$ draw.

Preparing the Epi Component for Analysis

In contrast to the LEWIE component, values for the variables I_h and Y in the Epi component are not sourced from survey data or model assumptions. Instead, values for the variables I_h and Y are "free" variables that are solved for simultaneously along with values in the LEWIE component, which ensures that the EEL model is in equilibrium within and between component models. For equilibrium to be reached in the EEL model, it needs to be the case that baseline household infection rates are consistent with the levels of effective labour supplied by households to the local economy, and that baseline levels of exposure time and aggregate output observed in the local economy correspond to the same baseline levels of infection. Because the Epi and LEWIE component variables are solved simultaneously, the baseline solution values for I_h and Y meet these conditions.

Preparing the Eco Component for Analysis

The baseline level of the fish stock is identified by assuming that the fish stock is at a steady state, which means that absent any changes to the values in the growth equation, the fish stock level does not change over time. With this assumption in place, the left side of (S7) is equal to zero, and

$$X_{SS} r_{fstock}(1 - \frac{X_{SS}}{K_{lake}}) = HARV_{baseline} \tag{S8}$$

where X_{SS} is the steady-state level of the fish stock and $HARV_{baseline}$ is the baseline level of harvest by the fishing sector of the local economy. Equation (S8) states that the level of natural growth of the fish stock is equal to the level of harvest by the fishing sector.

The value of $HARV_{baseline}$, which is obtained by jointly solving the Epi and LEWIE components, can be used to solve for X_{SS}. Because we do not solve the Eco component simultaneously with the LEWIE and Epi components, one more adjustment is necessary. As observed in (S2), the fish stock enters into the production function for the fishing sector as an input. However, the level of the fish stock is unknown when solving for $HARV_0$ in a process that requires the production function for the fishing sector. To address this, the level of the fish stock is set equal to 1 when solving for the equilibrium values of the LEWIE and Epi components. The corresponding production function for the fishing sector is

$$QP_{fish} = A_{Restr,fish} \prod_{j=1}^{J} f_{j,fish}^{\delta_j} 1^{\delta_{fstock}} \ . \tag{S9}$$

where $A_{Restr,fish}$ is the fishing sector TFP shift parameter when $X = 1$. Since $X_{SS} \neq 1$, the value of $A_{Restr,fish}$ obtained by jointly solving the Epi and LEWIE components is *incorrect*. With X_{SS} obtained as described above, the correct value of the TFP shift parameter can be found as follows. Comparing (S9) with the unrestricted equation (introduced above in section 2.7),

$$QP_{fish} = A_{fish} \prod_{j=1}^{J} f_{j,fish}^{\delta_j} X^{\delta_{fstock}} \tag{S2}$$

the right sides of both equations can be set equal to each other, with

$$A_{Restr,fish} = A_{fish} X^{\delta_{fstock}}$$

obtained after cancelling terms. The correct value of the TFP shift parameter for the fishing sector, A_{fish}, can therefore be found by dividing the restricted value of A_{Restr} obtained in the joint LEWIE-Epi solution process by the value of X_{SS} raised to the power of δ_{fstock}.

Solving the EEL Model

With the EEL model now in equilibrium within and between component models, we simulate different types of policy interventions. Each policy intervention is introduced by changing values for one of the parameters in the model. We consider both one-time and recurring interventions to the model by analysing the model over a 10-year period in annual time steps. The static nature of the LEWIE component means that the local economy reaches equilibrium every year. The differential equation for the Eco component is treated as a step equation, for which the "step forward" is in annual increments. This means that for year t, the solutions for the LEWIE component are found after adjusting the fishing sector's intermediate demand for fuel (which depends on the size of the fish stock in year t). The size of the fish stock in the next year can be calculated using the step-equation form of (S7), which includes the total harvest from year t, converted to kilograms. The solutions for the LEWIE component for each year include values for fishing labour time for each household, aggregate output, and output in the fishing sector, which then enter into the equations for the Epi and Eco components.

For the Epi component of the model, we assume that the disease dynamics evolve at a faster pace than the relatively slower dynamics of the local economy or population of the composite fish stock. Consequently, equilibrium in the Epi component can respond to changes in the LEWIE and Eco components, while adjustment in the Epi component occurs more quickly. With this assumption in place, the values for I_h and Y are at their steady-state (SS) levels and enter into the model for year t as

$$\gamma I_{h,SS} = \beta \tau_h^E \epsilon_h^E Y (1 - I_{h,SS}) \tag{S10}$$

$$\mu Y_{SS} = \chi (1 - Y_{SS}) \sum_{h=1}^{H} g_h \tau_h^C \epsilon_h^C I_{h,SS} \tag{S11}$$

after setting $\dot{I}_h = 0$ and $\dot{Y} = 0$ and rearranging terms. The values of $I_{h,SS}$ and Y_{SS} for year t are solved for jointly with the values for the LEWIE component.

Appendices

A1 Assigning Slope and Median Parameter Values for β and γ

Equations for $\phi_\beta(Z)$ and $\phi_\gamma(Z)$ are used to determine the relative weights for the assigned minimum and maximum values of the exposure rate, β, and mortality rate of the parasite in the human host, γ, respectively:

$$\phi_\beta(Z) = \frac{1}{1 + \exp\left(\frac{Z - Z_{med}}{Z_{slope}}\right)} \tag{S12}$$

$$\phi_\gamma(Z) = \frac{1}{1 + \exp\left(\frac{-(Z - Z_{med})}{Z_{slope}}\right)} \tag{S13}$$

We assign time-invariant values for the median and slope parameters for β and γ in this function using per-capita GDP for each country in 2017 [6]. We convert each country value to Ugandan shillings (UGX) using the exchange rate of 3,600 UGX to 1 USD and convert to logged values. We calculate a scaling factor by dividing the baseline value of aggregate output Z, converted to per capita, by Uganda's per capita GDP. We add the logged value of the scale factor to the logged value for each country, producing a log-normal distribution of GDP values, scaled to the local economy.

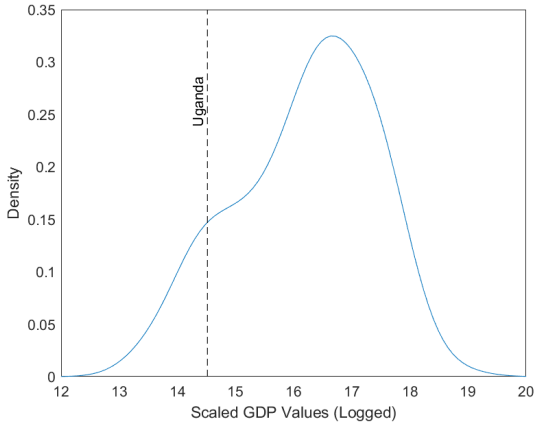


Figure S10: Distribution of GDP values.

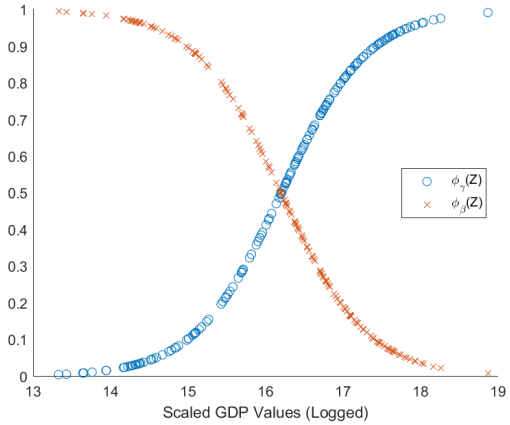


Figure S11: Scatter plots of $\phi_\gamma(Z)$ and $\phi_\beta(Z)$ against GDP values

Before any policy intervention is introduced to the model, the local economy for Bugala Island lies within the 12th percentile of this distribution. The distribution has a median value of 16.38, which we assign as the value of Z_{med}, and a standard deviation of 1.176. Empirically estimating the slope parameter Z_{slope} is not possible given the limitations of the data. We follow the approach used in Garchitorena *et al.*(2017) [4] by assigning a value for the slope parameter equal to 10% of the difference between the minimum and maximum values of the distribution. The steepness of the curve around the median value of the curve shown in Figure S11 depends on the value of the slope parameter.

A2 Constructing the Next-Generation Matrix (NGM) Used to Derive $\mathcal{R}_0$

We first consider a fixed, homogeneous population of N individuals that are either susceptible to (denoted as S) or are infected with (denoted as I) a generic disease. Population shares for susceptible and infected individuals are written as $s = {}^{S}/_{N}$ and $i = {}^{I}/_{N} = 1 - s$, respectively. The disease has the following transmission characteristics:

- The effective contact rate, denoted as b, is the rate at which the disease is passed from an infected individual to a susceptible individual. This rate is a product of the number of infections per contact and the number of contacts per time period.
- The rate of recovery, denoted as a, is the rate at which an infected individual becomes susceptible to the disease. The inverse of a is the mortality rate for the disease and can be thought of as the rate at which the disease "dies off" in an infected individual.

We can write the equation that represents how the population share of infected individuals changes over time as

$$\frac{di}{dt} = bsi - ai \tag{S14}$$

and use this equation to find the steady-state values of i; that is, the values for which i does not change over time, which is also when (S14) is equal to zero and thus can be solved as a quadratic equation with two roots. At the disease-free equilibrium (DFE), the entire population is free of infection

$$i_{DFE} = 0 \tag{S15}$$

while at the endemic equilibrium (EE), a share of the population is infected according to the equation

$$i_{EE} = 1 - \frac{a}{b} \tag{S16}$$

while the remaining share of the population is susceptible to the disease.

The basic reproduction number, or $\mathcal{R}_0$, is a dimensionless value that is defined as the average number of secondary cases arising from one new infection when the entire population is susceptible [15] The value of $\mathcal{R}_0$ tells us which of the two equilibria—the DFE or the EE—is stable. If $R_0 < 1$, the DFE is stable and i will decline toward zero over time. On the other hand, the EE is stable whenever $R_0 > 1$.

To identify the expression for the $\mathcal{R}_0$ for this generic disease in the case of a homogeneous population, we take the derivative of (S14) with respect to i around the DFE (i.e., where $i = 0$ and thus $s = 1$, which is the case when the entire population is susceptible). We obtain

$$\frac{\partial \dot{i}}{\partial i}\Big|_{i=0} = b - a \tag{S17}$$

which we assume is positive, because we know that $\mathcal{R}_0$ is a positive number (i.e., it is not possible to have a negative number of secondary cases or a negative infection rate). We can therefore write the $\mathcal{R}_0$ for equation (S14) as

$$R_0 = \frac{b}{a} \tag{S18}$$

which we can use to more formally restate the relationship between $\mathcal{R}_0$ and the two equilibria for the system of disease dynamics. Whenever $a > b$, $\mathcal{R}_0$ will be less than 1 and the DFE will be stable, with i trending toward zero over time and remaining at that level until the value of a or b changes. Whenever $b > a$, $\mathcal{R}_0$ will be greater than 1 and the EE will be stable, with i trending toward i_{EE} and remaining at that level until the value of either a or b changes.

A Heterogeneous Population

We now consider that the same population of N individuals can be divided into two closed subpopulation groups (i.e., a member of one subpopulation cannot transfer to the other subpopulation), and the two groups are heterogeneous such that the two rates identified above vary across the two groups (i.e., $a_1 \neq a_2$ and $b_1 \neq b_2$). The system of equations

$$\frac{di_1}{dt} = b_1 s_1 i_1 - a_1 i_1 \tag{S19}$$

$$\frac{di_2}{dt} = b_2 s_2 i_2 - a_2 i_2 \tag{S20}$$

represents transmission characteristics for the same generic disease introduced above. This system of equations has two equilibria that can be found by setting the left sides of (S19) and (S19) equal to zero and solving for the roots of the resulting quadratic equations. The first equilibrium point is the disease-free equilibrium (DFE), which exists when infection rates for both subpopulation groups are equal to zero. The endemic equilibrium (EE) exists where

$$i_{1,EE} = 1 - \frac{a_1}{b_1}$$

$$i_{2,EE} = 1 - \frac{a_2}{b_2}$$

and the remaining population is susceptible to the disease. For such scenarios with heterogeneous groups within a population, the $\mathcal{R}_0$ for the disease is the spectral radius (i.e., the "dominant" eigenvalue, or eigenvalue that is largest in absolute value) of the next-generation matrix (NGM) constructed using the above system of equations. The NGM consists of a "numerator" matrix, which we denote as $\mathcal{F}$, and a "denominator" matrix, which we denote as $\mathcal{V}$. To construct these matrices, we first divide the above system in two by separating terms associated with transition into each group with the numerator matrix and terms associated with transition out of each group with the denominator matrix. Specifically, we can rewrite (S19) and (S20) as

$$\frac{di_1}{dt} = \underbrace{b_1 s_1 i_1}_{F_1} - \underbrace{a_1 i_1}_{V_1} \tag{S21}$$

$$\frac{di_2}{dt} = \underbrace{b_2 s_2 i_2}_{F_2} - \underbrace{a_2 i_2}_{V_2} \tag{S22}$$

where

$$F = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix} \tag{S23}$$

and

$$V = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} . \tag{S24}$$

which can then be linearised around the DFE as follows. The numerator matrix for the NGM is a square matrix of partial derivatives that can be written as

$$\mathcal{F} = \begin{bmatrix} \frac{\partial F_1(i_1^*, i_2^*)}{\partial i_1} & \frac{\partial F_1(i_1^*, i_2^*)}{\partial i_2} \\ \frac{\partial F_2(i_1^*, i_2^*)}{\partial i_1} & \frac{\partial F_2(i_1^*, i_2^*)}{\partial i_2} \end{bmatrix} = \begin{bmatrix} b_1 & 0 \\ 0 & b_2 \end{bmatrix}$$

where each partial derivative term is evaluated by setting infection rates equal to zero (denoted using asterisks). Similarly, the denominator matrix for the NGM can be written as

$$\mathcal{V} = \begin{bmatrix} \frac{\partial V_1(i_1^*, i_2^*)}{\partial i_1} & \frac{\partial V_1(i_1^*, i_2^*)}{\partial i_2} \\ \frac{\partial V_2(i_1^*, i_2^*)}{\partial i_1} & \frac{\partial V_2(i_1^*, i_2^*)}{\partial i_2} \end{bmatrix} = \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix}$$

where the inverse matrix $\mathcal{V}^{-1}$ exists and can be written as

$$\mathcal{V}^{-1} = \begin{bmatrix} \frac{1}{a_1} & 0 \\ 0 & \frac{1}{a_2} \end{bmatrix}.$$

The NGM for the system of equations represented by (S19) and (S20) is

$$\mathcal{F}\mathcal{V}^{-1} = \begin{bmatrix} \frac{b_1}{a_1} & 0 \\ 0 & \frac{b_2}{a_2} \end{bmatrix} \tag{S25}$$

which leaves us with the final step of finding the spectral radius, or dominant eigenvalue, of the NGM. Consequently, we can write the $\mathcal{R}_0$ for this system as

$$R_0 = \rho(\mathcal{F}\mathcal{V}^{-1}) = \max\left\{\left|\frac{b_1}{a_1}\right|, \left|\frac{b_2}{a_2}\right|\right\} \tag{S26}$$

which is as much as we can conclude until we assign values to the system parameters.

An $\mathcal{R}_0$ for the EEL Model

State Equations

The state equation for household h's infection rate can be written as

$$\dot{I}_h = \beta \tau_h^E \epsilon_h^E Y(1 - I_h) - \gamma I_h \tag{S27}$$

where I_h denotes the infection rate for household h and Y denotes the infection rate for the host snail population in the single water source. For household h, transition into the infected classification is determined by: Y, the susceptible portion of the household $(1 - I_h)$, the snail-to-human transmission rate parameter β, a household-specific measure of exposure time τ_h^E, and a household-specific exposure-risk parameter ϵ_h^E.[3] Transition out of the infected classification is determined by the parameter γ, which is defined as the mortality rate of the parasite in the human host.

The state equation for the infected share of the snail population can be written as

$$\dot{Y} = \chi(1 - Y) \sum_{h=1}^{H} g_h \tau_h^C \epsilon_h^C I_h - \mu Y \tag{S28}$$

where the transition of the snail population from susceptible to infected is determined by: the size of the susceptible proportion $(1 - Y)$; the human-to-snail transmission rate parameter χ; and a weighted sum of the infection rates of the H households. The weighted sum of household infection rates in (S28) is calculated using household population shares, g_h, household-specific measures of contamination risk, ϵ_h^C, and a household-specific measure of contamination time τ_h^C.[3] For the snail population, transition out of the infected classification is determined by the parameter μ, which is defined as the mortality rate of the parasite in the snail host.

Equilibria and R_0

The system of disease dynamics described by (S27) and (S28) admits two steady-state equilibrium points. The Disease-Free Equilibrium (DFE) is characterised by zero values of the state variables I_h and Y. The

Endemic Equilibrium (EE) is characterised by positive values of the same state variables. The stability of either equilibrium point can be determined by the value of the community-level reproduction number, R_0, a dimensionless value that is defined as the average number of secondary cases across subpopulation groups arising from one new infection when the entire population is susceptible.[15] Since R_0 is defined in terms of an average across the different groups in a population, R_0 is unique to the system of disease dynamics for the local economy and common across the H subpopulation groups.

Following Diekmann *et al.*(2009) [15] and the applications thereof,[4, 16] we calculate R_0 as the dominant eigenvalue of the next-generation matrix for the above system of disease dynamics. For a local economy with H households, this matrix can be written as $\mathcal{F}\mathcal{V}^{-1}$, where

$$\mathcal{F} = \begin{bmatrix} 0 & \dots & 0 & \beta\tau_1^E\epsilon_1^E \\ 0 & \ddots & 0 & \vdots \\ 0 & 0 & 0 & \beta\tau_H^E\epsilon_H^E \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathcal{V} = \begin{bmatrix} -\gamma & \dots & 0 & 0 \\ 0 & \ddots & 0 & \vdots \\ 0 & 0 & -\gamma & 0 \\ \chi g_1\tau_1^C\epsilon_1^C & \dots & \chi g_H\tau_H^C\epsilon_H^C & -\mu \end{bmatrix}.$$

The R_0 of the system described by (S27) and (S28) can be written as

$$R_0 = \frac{\beta\chi\sum_{h=1}^{H} g_h\tau_h^E\tau_h^C\epsilon_h^E\epsilon_h^C}{\mu\gamma}. \tag{S29}$$

When $R_0 < 1$, the DFE is stable and the values for the state variables I_h and Y will trend toward zero. When $R_0 > 1$, the EE is stable and the state variables I_h and Y will trend toward their respective EE values as determined by the system's parameter values.

A3 Equations of the EEL Model

The tables below contain the equations that comprise each component of the EEL model. The Epi component contains 9 equations, the Eco component contains 1 equation, and the LEWIE component contains 350 equations. Each equation corresponds to a variable whose value is identified in the solution process.

Table S14: Equations and variables for the Epi component of the EEL model

Equation Type (Number of Eqs.)	Equation Statement	Variable
State Equation: Household Infection Status (4)	$\dot{I}_h = \beta(Z)\tau_h\epsilon_h Y(1 - I_h) - \gamma(Z)I_h$	I_h
State Equation: Snail Infection Status (1)	$\dot{Y} = \chi(1 - Y)\sum_{h=1}^{H} g_h\tau_h\epsilon_h I_h - \mu Y$	Y
Household Exposure Time (4)	$\tau_h = \bar{\tau}_h + \frac{L_{h,fish}}{L_{h,fish,baseline}}$	τ_h

Table S15: Equations and variables for the Eco component of the EEL model

Equation Type (Number of Eqs.)	Equation Statement	Variable
State Equation: Fish Stock (1)	$\dot{X} = Xr_{fstock}(1 - \frac{X}{K_{lake}}) - HARV$	X

Table S16: Equations and variables for the LEWIE component of the EEL model

Equation Type (Number of Eqs.)	Equation Statement
Value-Added Prices (24)	$PVA_{g,h} = PVA(p_g; idsh_{g,gg,h})$
Value-Added Output (24)	$QVA_{g,h} = QVA\left(\begin{array}{l} FD_{g,f,h}; fshare_{g,f,h}, \\ pshift_{g,h}, stock_g, stockbeta_g \end{array} \right)$
Factor Demand (96)	$FD_{g,f,h} = FD\left(\begin{array}{l} r_{g,f,h}, w_f, PVA_{g,h}, QP_{g,h}; \\ theta_{f,h}, stockbeta_{g,h}, fshare_{g,f,h} \end{array} \right)$
Output (6)	$QP_{g,h} = QP(QVA_{g,h}, vash_{g,h})$
Intermediate Demands (96)	$ID_{g,f,h} = ID(QP_{g,h}, idsh_{g,f,h})$
Household Consumption (24)	$QC_{g,h} = QC\left(\begin{array}{l} p_g, HHEXP_h, trout_h, sav_h, expout_h; \\ util_sh_{g,h}, good_elas_h \end{array} \right)$
Household Income (4)	$HHEXP_h = HHEXP\left(\begin{array}{l} r_{g,f,h}, fd_{g,f,h}, w_{f,h}, \\ efflabsup_h; exinc_h \end{array} \right)$
Household Marketed Surplus (24)	$HMS_h = HMS(QP_{g,h}, QC_{g,h}, ID_{gg,g,h})$
Economywide Marketed Surplus (6)	$VMS = VMS(p_g, HMS_{g,h})$
Household Factor Marketed Surplus (8)	$HFMS_h = HFMS(HFSUP_{f,h}, FD_{g,f,h})$
Economywide Factor Marketed Surplus (2)	$VFMS = VFMS(HFMS_h)$
Transfers out (4)	$TROUT_h = TROUT(HHEXP_h; trout_sh_h)$
Household Savings (4)	$SAV_h = SAV(HHEXP_h; sav_sh_h)$
Exogenous Expenditures of the Household (4)	$EXPROC_h = EXPROC(HHEXP_h; exproc_sh_h)$
Output for Composite Goods (Armington Function) (1)	$QP_Comp = QP_Comp\left(\begin{array}{l} QP_{g,h}, Imports_g; \\ armg_shift_g, delta_g, rho_g \end{array} \right)$
Prices for Composite Goods (Armington Function) (1)	$P_Comp = P_Comp(QP_{g,h}, p_g; delta_g, rho_g)$

Continued on next page

Table S16 – continued from previous page

Equation Type (Number of Eqs.)	Equation Statement
Imports (1)	$Imports_g = Imports(QP_Comp, QP_{g,h}, p_g, Imports_g)$
Household Labour Supply (4)	$HFSUP_h = HFSUP\begin{pmatrix} Labtime_B L_h, I_h, w_{f,h}; \\ \alpha, lab_elas_h \end{pmatrix}$
Consumer Price Index (4)	$CPI_h = CPI(p_g, QC_{g,h})$
Household Real Income (4)	$RY = RY(HHEXP_h, CPI_h)$

458

A4 Parameterizing the MDA Intervention

To model the impact of an MDA program targeted at reducing Schisto prevalence, we introduce the parameter γ_{mda} into the system of equations for the disease dynamics:

$$I_{h,t+1} - I_{h,t} = \beta_t \tau_{h,t} \epsilon_h Y_t (1 - I_{h,t}) - \gamma_t (1 + \gamma_{mda}) I_{h,t} \tag{S30}$$

$$Y_{t+1} - Y_t = \beta_t (1 - Y_t) \sum_{h=1}^{H} g_h \tau_{h,t} \epsilon_h I_{h,t} - \mu Y_t \tag{S31}$$

The term $\gamma_t(1+\gamma_{mda})$ represents the rate of transition out of the infection classification for the subpopulation group. We model γ_t as increasing in aggregate output with the following minimum and maximum values:

$$\gamma_{min} = 5.5 \times 10^{-4} \tag{S32}$$

$$\gamma_{max} = 5.5 \times 10^{-3} \tag{S33}$$

We assign a value for γ_{mda} that produces a decline in infection rates that is consistent with prior literature. We first identify a value for annual reduction rates in Schisto infection that can result from community-wide MDA programs in communities similar to the one under study here. King *et al.*(2020) [17] study the impact of annual community-based treatment for Schisto on infection rates in three countries: Kenya, Mozambique, and Tanzania. Evaluating changes in prevalence rates between year 1 and year 5 after program implementation, they find a decline in prevalence of approximately 74% (Kenya), 36% (Tanzania), and 55% (Mozambique). These changes are equivalent to annual reduction rates of 29% (Kenya), 11% (Tanzania), and 18% (Mozambique), and the simple average of these rates is 19.3%.

To solve for γ_{mda}, we convert the left side of (S30) to a percentage change by dividing both sides by I_h and set the left side of the equation equal 0.193, which is the annual reduction rate identified above. We solve for the desired value of γ_{mda} using baseline values of the Epi state variables I_h and Y and the parameters β, γ, and τ_h. Since the MDA treatment only targets the human population, we assume no changes in Y over the study period, which is consistent with results shown in Figure 7b from Castonguay *et al.*(2020) [16]. We calculate a value of γ_{mda} equal to 0.295 that we use to model the MDA policy intervention in this study.

A5 Robustness Check: Low and High Values of α

Results presented in the main text assume that a unit of labour time is 15% less productive due to Schisto infection. The relationship between Schisto and labour productivity is one of three links between the public health and economic domains of the human-natural environment. Given the range of estimates from previous studies,[18, 19] we explore in this section the sensitivity of our results to changes in the value of α.

Schisto and Labour Effectiveness

To examine whether results are sensitive to changes in the value of α, we re-estimate the effects of the FMP, TFP, and MDA policy interventions for the minimum ($\alpha = 0.05$) [18] and maximum ($\alpha = 0.3$) [19] values. The importance of the parameter α in the EEL model can be seen by reflecting on the equation

$$L_{h,t} = \frac{E_{h,t}}{(1 - \alpha I_{h,t})} \tag{S34}$$

which represents the relationship between the household supply of effective labour and its supply of labour time in year t. For a given household infection rate, I_h, the parameter α determines the difference between the two measures of labour supply. Furthermore, the measure of fishing labour time supplied by the household also appears in the equation for exposure time in year t for household h

$$\tau_{h,t} = \bar{\tau}_h + \frac{L_{h,fish,t}}{L_{h,fish,baseline}} \tag{S35}$$

where $\bar{\tau}_h$ is constant over time and accounts for background household activities that are correlated with exposure to Schisto, such as time spent collecting water. The time-varying component of (S35) includes the amount of fishing labour time supplied by the household, $L_{fish,h,t}$. Together, (S34) and (S35) imply that when we re-estimate the effects of the three policy interventions–MDA, FMP, and TFP–using low (5%) and high (30%) values of α, we should expect to see differences in changes in exposure time across the two values of α.

Recalling the equation for $\mathcal{R}_0$ and the state equation for the household infection rate, I_h,

$$R_0 = \frac{\beta^2 \sum_{h=1}^{H} g_h \tau_h^2 \epsilon_h^2}{\mu\gamma} \tag{S36}$$

$$\dot{I}_h = \beta \tau_h \epsilon_h Y (1 - I_h) - \gamma I_h \tag{S37}$$

we observe that $\mathcal{R}_0$ is increasing in exposure time, since the measure of exposure time, τ_h, appears in the numerator of the expression for $\mathcal{R}_0$. Similarly, we also observe that $\dot{I}_h$ is increasing in τ_h. It follows that $\mathcal{R}_0$ and $\dot{I}_h$ are increasing in α, even though α does not appear explicitly in the equation for $\mathcal{R}_0$. Consequently, we also anticipate seeing variation in results for $\mathcal{R}_0$ and household infection rates across low and high values of α.

Results

For all three policy interventions, changes in exposure time vary across low and high values of α. We also observe that changes in $\mathcal{R}_0$ vary, as anticipated due to the implicit relationship between $\mathcal{R}_0$ and α. However, these differences do not carry over for aggregate output, as the changes in aggregate output across policy interventions are robust to values of α.

Outcomes in the ecological domain are robust to changes in the value of α (see Table S17).

Variation in changes in exposure time translate into differences for household infection rates, although not enough to affect the conclusions reached in our main analysis.

Differences in changes in exposure time spill partially into the economic domain. Changes in household incomes vary slightly, which is a consequence of the relationship between α, labour time, and effective labour (see (S34)). Specifically, a larger value of α results in a smaller amount of effective labour, for a given amount of labour time. Nevertheless, the differences in household incomes observed do not affect the conclusions reached in our main analysis.

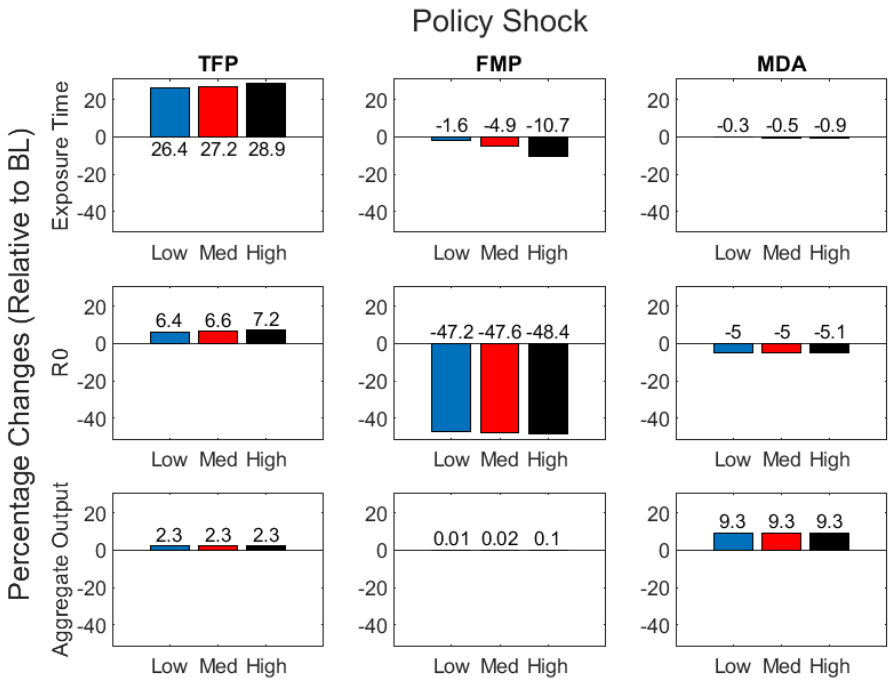


Figure S12: Year-10 percentage changes (relative to baseline) in exposure time, $\mathcal{R}_0$, and aggregate output across policy interventions for low (5%), med (15%), and high (30%) values of α. Underlying data provided in Table S17.

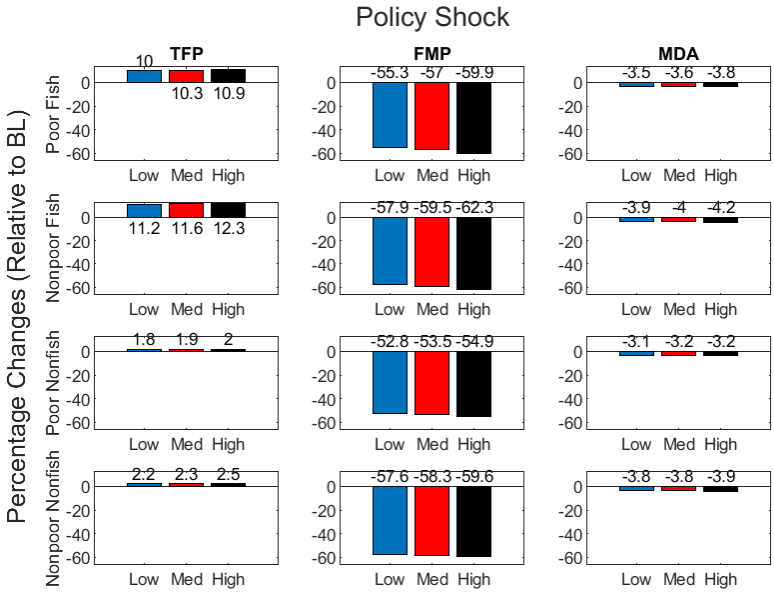


Figure S13: Year-10 percentage changes (relative to baseline) in household infection rates across policy interventions for low (5%), med (15%), and high (30%) values of α. Underlying data provided in Table S17.

Table S17: Percentage Changes (Relative to Baseline) in Outcomes

Outcome		FMP		MDA		TFP	
		Low	High	Low	High	Low	High
$\mathcal{R}_0$		6.4	7.2	-47.2	-48.4	-5.0	-5.1
Fish Stock		42.6	42.6	0.0	-0.1	0.3	0.3
Aggregate Output		2.3	2.3	0.0	0.1	9.3	9.3
Fish Harvest		12.6	12.6	0.0	0.0	0.1	0.1
Exposure Time (τ_h)		26.4	28.9	-1.6	-10.7	-0.3	-0.9
Household Income	Poor, Fishing	19.4	18.8	0.0	0.3	7.8	7.9
	Nonpoor, Fishing	8.9	8.7	0.0	0.0	4.0	4.1
	Poor, Nonfishing	2.2	2.3	0.0	0.3	8.5	8.5
	Nonpoor, Nonfishing	0.9	0.9	0.0	-0.1	5.5	5.5
Infection Rates	Poor, Fishing	10.0	10.9	-55.3	-59.9	-3.5	-3.8
	Nonpoor, Fishing	11.2	12.3	-57.9	-62.2	-3.9	-4.2
	Poor, Nonfishing	1.8	2.0	-52.8	-54.9	-3.1	-3.2
	Nonpoor, Nonfishing	2.2	2.5	-57.5	-59.6	-3.8	-3.9

Notes: Low value of $\alpha = 0.05$, high value of $\alpha = 0.3$.

A6 Robustness Check: Alternative Specifications of Labour Supply Elasticity

Results presented in the main text assume a highly elastic local labour supply ($\eta = 100$), consistent with the high rates of unemployment and underemployment documented across rural Africa and, more specifically, in Kalangala District [2, 1]. Because η in this setting is not directly estimated from our survey data, we examine in this section the sensitivity of our results presented in the main text, re-estimating the standalone MDA, FMP, and TFP interventions under two alternative specifications.

Two Alternative Specifications

The first alternative specification sets $\eta = 1$ and represents a setting in which the quantity of labour supplied responds much less readily to wage changes than in our preferred specification. The second alternative specification allows the elasticity to vary with aggregate infection prevalence, reflecting the idea that a population with a greater infection burden may be less able to reallocate labour in response to economic incentives. We specify this second alternative as

$$\eta_t = \max\left(1,\ 100 - 100\sum_{h=1}^{H} s_h I_{h,t-1}\right) \tag{S38}$$

where s_h is household h's population share and $\sum_h s_h I_{h,t-1}$ is the population-weighted aggregate infection rate entering year t, bounded so that $\eta_t \in [1, 100]$. We use the aggregate infection rate entering year t (i.e., determined by year $t-1$'s outcomes) because that is the value already available to the household's labour supply decision at the time it is made.

Results

Tables S18, S19, and S20 report Year 1 and Year 10 percentage changes from baseline for the MDA, FMP, and TFP interventions, respectively, under all three elasticity specifications, for the RoW trade scenario presented in Table 1 of the main text.

Across all three interventions, we observe that the direction of every outcome is unchanged across the three elasticity specifications. The public health outcomes, $\mathcal{R}_0$ and infection prevalence, are the least sensitive to the labour supply elasticity, since these results are driven primarily by exposure time and disease transmission parameters that do not depend directly on η. The small differences that do appear arise through the modest reallocation of labour governed by the labor supply elasticity.

The economic and ecological outcomes (fishing labour time, aggregate output, fish stock, and harvest) show larger sensitivity to the elasticity assumption, which is expected given that these outcomes are determined directly by how readily labour reallocates in response to changing wages. The largest differences arise under $\eta = 1$, the most restrictive of the two alternative specifications, and are most pronounced for the TFP intervention's effect on the fish stock (0.3% under our preferred specification versus 3.1% under $\eta = 1$) and for MDA's effect on aggregate output and the fish stock, where results move from a negligible year-10 effect to changes near or above one percent. Nevertheless, these differences do not change which direction each outcome moves, nor the relative ordering of effects across interventions.

Results under the declining-elasticity specification track closely with our preferred specification throughout. For all three interventions considered here, infection prevalence is either falling steadily (MDA, TFP) or does not rise enough to meaningfully depress η_t (FMP), so the labour supply elasticity recovers toward its ceiling of 100 within the first few years of the simulation, and the results converge with those of our preferred specification well before Year 10.

Taken together, these results indicate that the qualitative conclusions reached in the main text, including the direction and, in most cases, the approximate magnitude of each intervention's effects across the public health, economic, and ecological domains, are robust to the choice of labour supply elasticity.

Table S18: Percentage Changes (Relative to Baseline) in Outcomes for the MDA Intervention Under Alternative Labour Supply Elasticity Specifications. Fishing sector output prices are *exogenous* to changes in the local economy.

Outcome	$\eta = 100$ (main text)		$\eta = 1$		η declining in infection	
	Year 1	Year 10	Year 1	Year 10	Year 1	Year 10
$\mathcal{R}_0$	-22.78	-47.64	-22.78	-47.71	-22.78	-47.64
Infection Prevalence	-19.59	-56.41	-19.59	-56.56	-19.59	-56.41
Fishing Labour Time	0.00	-4.94	0.00	-4.26	0.00	-4.94
Aggregate Output	0.00	0.02	0.00	0.95	0.00	0.03
Fish Stock[1]	0.00	-0.04	0.00	-1.50	0.00	-0.05
Harvest (kg)	0.00	-0.02	0.00	-0.69	0.00	-0.02

[1]Fish stock values are reported as changes in fraction of carrying capacity.

Table S19: Percentage Changes (Relative to Baseline) in Outcomes for the FMP Intervention Under Alternative Labour Supply Elasticity Specifications. Fishing sector output prices are *exogenous* to changes in the local economy.

Outcome	$\eta = 100$ (main text)		$\eta = 1$		η declining in infection	
	Year 1	Year 10	Year 1	Year 10	Year 1	Year 10
$\mathcal{R}_0$	-2.88	6.63	-2.97	6.50	-2.89	6.62
Infection Prevalence	-2.23	3.50	-2.23	3.51	-2.23	3.50
Fishing Labour Time	-24.65	27.21	-22.34	25.06	-24.58	27.14
Aggregate Output	-5.97	2.34	-4.77	1.36	-5.94	2.30
Fish Stock[1]	17.67	42.57	17.11	44.01	17.65	42.62
Harvest (kg)	-24.88	12.63	-24.10	12.70	-24.86	12.63

[1]Fish stock values are reported as changes in fraction of carrying capacity.

Table S20: Percentage Changes (Relative to Baseline) in Outcomes for the TFP Intervention Under Alternative Labour Supply Elasticity Specifications. Fishing sector output prices are *exogenous* to changes in the local economy.

Outcome	$\eta = 100$ (main text)		$\eta = 1$		η declining in infection	
	Year 1	Year 10	Year 1	Year 10	Year 1	Year 10
$\mathcal{R}_0$	-0.46	-5.00	-0.49	-4.38	-0.47	-4.99
Infection Prevalence	-0.32	-3.55	-0.33	-3.08	-0.32	-3.54
Fishing Labour Time	-0.06	-0.51	-0.65	-2.86	-0.08	-0.58
Aggregate Output	0.85	9.30	0.60	6.98	0.84	9.24
Fish Stock[1]	0.01	0.26	0.16	3.06	0.02	0.34
Harvest (kg)	-0.02	0.07	-0.22	0.84	-0.03	0.10

[1]Fish stock values are reported as changes in fraction of carrying capacity.

A7 Robustness Check: Lagged Response of Labour Effectiveness to Infection

Results presented in the main text link the effectiveness of household labour in year t to the equilibrium infection prevalence reached in year $t-1$, so that a change in infection reaches labour productivity within one year. Morbidity from schistosomiasis follows cumulative worm burden, however, and adult worms live an average of three to ten years in their human hosts [20]. Prevalence is therefore a proxy for the worm burden that drives morbidity, and labour productivity in the model may respond more quickly to changes in exposure than it would if burden accumulated and cleared over several years. Infection intensity also varies across infected individuals, and chronic infection can lead to worse outcomes [21]. We examine in this section the sensitivity of our results presented in the main text to this simplification, re-estimating the standalone FMP and TFP interventions with a lagged response of labour effectiveness to infection.

Lagged Infection Burden

Let $I_{h,t}$ denote the equilibrium infection prevalence of household h in year t. We allow the infection burden that reduces labour effectiveness to adjust only partially toward this equilibrium value each year,

$$B_{h,t} = \lambda I_{h,t} + (1-\lambda) B_{h,t-1} \tag{S39}$$

where $\lambda \in (0,1]$ is the share of the gap between the burden and the equilibrium prevalence that is closed each year. The burden carried into year $t+1$ replaces the infection rate in the equation for effective labour, so that $E_{h,t+1} = L_{h,t+1}\left(1-\alpha B_{h,t}\right)$. The model presented in the main text is the special case in which $\lambda = 1$. The initial burden, $B_{h,0}$, is the baseline infection prevalence, so Year 1 results are identical across all values of λ.

An adult worm lifespan of L years implies $\lambda = 1 - e^{-1/L}$, which is roughly 0.1 for a lifespan of ten years and 0.3 for a lifespan of three years. We therefore report results for λ equal to 1, 0.5, 0.25, and 0.1. We apply this rule to the FMP and TFP interventions only. Praziquantel acts against adult worms and its effect is short in duration (days) relative to the annual time step of the model [22], so the MDA intervention reduces the worm burden directly and is unchanged. Reported infection prevalence and $\mathcal{R}_0$ remain the equilibrium values, and the sensitivity isolates the lag between a change in infection and its consequences for labour productivity.

Results

Tables S21 and S22 report Year 5 and Year 10 percentage changes from baseline for the FMP and TFP interventions under the RoW trade scenario presented in Table 1 of the main text, and Tables S23 and S24 report the corresponding results under the Local trade scenario presented in Table 2.

Across both interventions and both trade scenarios, we observe that the direction of every outcome is unchanged across values of λ. The largest differences arise under the FMP, where the rebound in fishing labour time and infection prevalence is slightly smaller in the later years of the simulation. For example, with $\lambda = 0.1$ in the RoW scenario, fishing labour time is 22.55% above baseline in Year 5 compared with 23.39% in the main text, and $\mathcal{R}_0$ is 5.50% above baseline compared with 5.76%. The corresponding Year 5 values in the Local scenario are 13.31% and 3.62% compared with 13.73% and 3.74%. Household income responses differ by less than 0.3 percentage points. Under the TFP intervention, all outcomes differ by less than 0.2 percentage points.

Aggregate output, fish stock, and harvest are essentially unchanged under both interventions, since the lag affects these outcomes only indirectly through labour effectiveness. Taken together, these results indicate that the qualitative conclusions reached in the main text are robust to the rate at which labour effectiveness adjusts to changes in infection.

Table S21: Percentage Changes (Relative to Baseline) in Outcomes for the FMP Intervention Under Alternative Rates of Adjustment of the Infection Burden. Fishing sector output prices are *exogenous* to changes in the local economy.

Outcome	$\lambda = 1$ (main text)		$\lambda = 0.5$		$\lambda = 0.25$		$\lambda = 0.1$	
	Year 5	Year 10	Year 5	Year 10	Year 5	Year 10	Year 5	Year 10
$\mathcal{R}_0$	5.76	6.63	5.63	6.61	5.54	6.54	5.50	6.40
Infection Prevalence	3.13	3.50	3.07	3.49	3.02	3.46	3.00	3.39
Fishing Labour Time	23.39	27.21	22.99	27.17	22.69	26.92	22.55	26.48
Aggregate Output	1.75	2.34	1.76	2.34	1.76	2.34	1.76	2.34
Fish Stock	41.16	42.57	41.16	42.57	41.16	42.57	41.16	42.57
Harvest (kg)	10.00	12.63	10.00	12.63	10.01	12.63	10.01	12.63
Income, Poor Fishing Households	16.68	19.20	16.78	19.21	16.86	19.27	16.89	19.38

Table S22: Percentage Changes (Relative to Baseline) in Outcomes for the TFP Intervention Under Alternative Rates of Adjustment of the Infection Burden. Fishing sector output prices are *exogenous* to changes in the local economy.

Outcome	$\lambda = 1$ (main text)		$\lambda = 0.5$		$\lambda = 0.25$		$\lambda = 0.1$	
	Year 5	Year 10	Year 5	Year 10	Year 5	Year 10	Year 5	Year 10
$\mathcal{R}_0$	-2.41	-5.00	-2.40	-4.99	-2.39	-4.98	-2.38	-4.96
Infection Prevalence	-1.64	-3.55	-1.64	-3.55	-1.63	-3.54	-1.63	-3.52
Fishing Labour Time	-0.27	-0.51	-0.24	-0.47	-0.21	-0.41	-0.18	-0.32
Aggregate Output	4.44	9.30	4.44	9.30	4.44	9.30	4.44	9.30
Fish Stock	0.11	0.26	0.11	0.26	0.11	0.26	0.11	0.26
Harvest (kg)	0.00	0.07	0.00	0.07	0.00	0.07	0.00	0.07
Income, Poor Fishing Households	3.71	7.83	3.71	7.83	3.71	7.83	3.71	7.83

Table S23: Percentage Changes (Relative to Baseline) in Outcomes for the FMP Intervention Under Alternative Rates of Adjustment of the Infection Burden. Fishing sector output prices are *endogenous* to changes in the local economy.

Outcome	$\lambda = 1$ (main text)		$\lambda = 0.5$		$\lambda = 0.25$		$\lambda = 0.1$	
	Year 5	Year 10	Year 5	Year 10	Year 5	Year 10	Year 5	Year 10
$\mathcal{R}_0$	3.74	4.55	3.70	4.54	3.66	4.50	3.62	4.42
Infection Prevalence	2.19	2.59	2.17	2.58	2.15	2.56	2.13	2.52
Fishing Labour Time	13.73	17.47	13.60	17.44	13.45	17.31	13.31	17.06
Aggregate Output	0.15	0.69	0.15	0.69	0.15	0.69	0.15	0.69
Fish Stock	41.79	48.51	41.79	48.51	41.79	48.51	41.79	48.50
Harvest (kg)	6.08	12.45	6.08	12.45	6.08	12.45	6.08	12.45
Income, Poor Fishing Households	10.09	13.10	10.12	13.11	10.16	13.14	10.20	13.21

Table S24: Percentage Changes (Relative to Baseline) in Outcomes for the TFP Intervention Under Alternative Rates of Adjustment of the Infection Burden. Fishing sector output prices are *endogenous* to changes in the local economy.

Outcome	$\lambda = 1$ (main text)		$\lambda = 0.5$		$\lambda = 0.25$		$\lambda = 0.1$	
	Year 5	Year 10	Year 5	Year 10	Year 5	Year 10	Year 5	Year 10
$\mathcal{R}_0$	-2.11	-4.56	-2.10	-4.56	-2.10	-4.54	-2.09	-4.53
Infection Prevalence	-1.46	-3.17	-1.46	-3.17	-1.46	-3.16	-1.45	-3.15
Fishing Labour Time	1.68	2.73	1.69	2.76	1.71	2.80	1.73	2.86
Aggregate Output	4.83	10.03	4.83	10.03	4.83	10.03	4.83	10.03
Fish Stock	-1.19	-3.23	-1.19	-3.23	-1.19	-3.23	-1.19	-3.23
Harvest (kg)	0.05	-0.83	0.05	-0.83	0.05	-0.83	0.05	-0.83
Income, Poor Fishing Households	4.98	9.81	4.99	9.81	4.99	9.81	4.99	9.82

A8 Robustness Check: Net Migration in Response to Household Income

In the main text, we hold the population of each household type fixed over the ten-year simulation. Migration that is the same in the baseline and in the policy scenarios cancels in deviations from baseline, but migration that responds to the policy interventions does not. We examine in this section how the results presented in the main text change when we relax the fixed population assumption and allow net migration into each household type to respond to that type's income per capita.

Population Index and Migration Rule

Let $P_{h,t}$ denote the population index of household type h in year t, where $P_{h,t} = 1$ is the baseline population. Each representative household stands for a number of actual households and their members, and the index scales that number, and with it the baseline labour time that the household type supplies to the local economy. We assume that migrants bring labour only, so each household type's capital and land remain fixed and are shared among more members as the population grows and among fewer as it declines. Population enters the disease dynamics in two places. Output per capita, which determines β and γ, and the population shares used to weight household types in $\mathcal{R}_0$ and in aggregate infection are computed using the current number of people represented. Exposure time is defined as in the main text, as the ratio of the household's fishing labour time to its baseline value, so migration changes exposure only if it changes fishing labour time.

All four household types adjust according to

$$P_{h,t+1} = P_{h,t}\, S_{h,t}^{\eta} \tag{S40}$$

where $S_{h,t}$ is income per capita of household type h in year t relative to its baseline value, and η is the annual elasticity of the population of a household type with respect to its own income per capita. For example, the fish stock affects migration directly through the income of households that fish, and the TFP intervention affects migration directly through the income generated in the oil palm sector. Because the model also captures spillovers into other sectors through local demand and prices, changes in those sectors can affect the income of other household types and therefore their migration. A 1% rise in income per capita raises the population of a household type by η% the following year, and by roughly 10η% over the ten-year horizon if the gap persisted. In practice, entrants dilute the income per capita of the household type, which lowers $S_{h,t}$ and slows later migration. We interpret the changes in the population index as net movement of people into or out of the study area, so migrants are not drawn from the other household types in the model.

The baseline without an intervention leaves $S_{h,t}$ at one, so migration is zero in the baseline and the reported effects remain deviations from baseline. The model presented in the main text is the special case in which $\eta = 0$. We treat the values $\eta = 0.025$ and 0.05 as illustrative, corresponding to ten-year elasticities of roughly 0.25 and 0.5, and we apply the same value to every household type because the evidence we draw on concerns regional income and not income by occupation. Econometric studies of rural-urban migration in Kenya, Tanzania, and other developing countries report elasticities of gross migration flows with respect to destination income of roughly 0.6 to 1.8 [23]. Because gross flows are a small share of the population each year, we expect the annual response of net population to income to be well below these elasticities. Evidence on the sign of the effect of income on migration is also mixed. In Kagera, Tanzania, temperature shocks that lower household consumption by 4.9% reduce the annual male long-term migration rate by 13% (from 5.8%), which suggests that higher income raises out-migration among liquidity-constrained households [24], and a study of international labour migration from rural Indonesia finds average income elasticities of roughly 0.1 for transitory and 1 for persistent shocks [25].

Results

Tables S25 to S30 report Year 5 and Year 10 percentage changes from baseline for the MDA, FMP, and TFP interventions under the RoW and Local trade scenarios. Year 1 results are identical to those in the main

text by construction. By Year 5 the differences from the main text are below 0.7 percentage points for every outcome, and they grow through Year 10 as population changes accumulate. Aggregate output and income are reported per capita because the population differs from the baseline under migration.

Migration does not change fishing effort in our specification. Fishing labour demand is determined by the fixed fishing capital stock, the fish stock, and the wage. Because labour supply is highly elastic, with an elasticity of 100, a change in population of $x\%$ changes the wage by only about $x/100\%$, so migration leaves fishing labour time, the fish stock, and harvest essentially unchanged, even under the FMP where the fishing populations rise most (Table S26). When we repeat the analysis with a labour supply elasticity of 1 (see Appendix A6), fishing labour time rises with the fishing population but by less than 0.5 percentage points, the fish stock changes by less than 0.3 percentage points, and the conclusions below are unchanged.

We find that the direction of every outcome is unchanged for both values of η with one exception that we describe below, and that most outcomes change little. The MDA intervention is essentially unaffected (Tables S25 and S28), because MDA changes household income very little. The TFP intervention raises income for all four household types, so it induces in-migration into every type, and the in-migration lowers income per capita and weakens the public health gains. In the RoW scenario with $\eta = 0.05$, for example, $\mathcal{R}_0$ falls by 4.16% rather than 5.00% (Table S27). The larger population lowers output per capita, which raises β and lowers γ.

The FMP is the intervention most affected. It raises the incomes of fishing households most, so their populations rise most, by about 4% for poor fishing households in Year 10 with $\eta = 0.05$ in the RoW scenario (Table S26). Fishing labour time, the fish stock, and harvest are essentially unchanged, so migration does not alter the rebound in fishing effort that drives the trade-off. The long-run increase in $\mathcal{R}_0$ is nevertheless larger, at 7.50% rather than 6.63% in that scenario. Decomposing the change, about 0.6 of the 0.9 percentage point increase comes from lower output per capita, which raises β and lowers γ, and about 0.3 comes from the shift in population weights toward fishing households, whose exposure time rises under the FMP. The larger population also lowers income per capita, and in the Local scenario it removes the small aggregate gain, with aggregate output per capita falling from 0.69% to -0.14% above baseline for $\eta = 0.05$ (Table S29). This is the only outcome in Tables S25 to S30 whose direction changes, and it is within 0.2 percentage points of zero.

Taken together, these results indicate that the qualitative conclusions reached in the main text are robust to allowing migration to respond to household income. Migration does not change the mechanism that drives the FMP trade-off. It lowers income per capita, and through lower output per capita and a shift in population weights it modestly raises $\mathcal{R}_0$, so the rebound in exposure under the FMP is somewhat larger than in the main text and the public health gains from TFP are somewhat smaller. Neither result is reversed.

Our migration rule has limitations that we state explicitly. Because migrants bring labour only, our estimates likely overstate the dilution of output per capita among households that receive migrants, and with it the increase in $\mathcal{R}_0$ that operates through β and γ, since migrants who bring capital would dilute output per capita less. Migrants who bring capital would also raise total fishing effort and pressure on the fish stock, which our specification cannot represent because fishing capital is fixed. We do not expect either limitation to reverse the direction of any result.

Table S25: Percentage Changes (Relative to Baseline) in Outcomes for the MDA Intervention Under Alternative Annual Elasticities of Population. Fishing sector output prices are *exogenous*.

Outcome		Main text		$\eta = 0.025$		$\eta = 0.05$	
		Year 5	Year 10	Year 5	Year 10	Year 5	Year 10
$\mathcal{R}_0$		-44.55	-47.64	-44.55	-47.63	-44.55	-47.63
Fish Stock		-0.03	-0.04	-0.03	-0.04	-0.03	-0.04
Aggregate Output per Capita		0.02	0.02	0.02	0.02	0.02	0.01
Fish Harvest		0.00	-0.02	0.00	-0.02	0.00	-0.02
Fishing Labour Time		-4.24	-4.94	-4.24	-4.94	-4.24	-4.94
Household Income per Capita	Poor, Fishing	0.07	0.08	0.07	0.07	0.07	0.06
	Nonpoor, Fishing	0.00	-0.01	0.00	-0.01	0.00	-0.01
	Poor, Nonfishing	0.08	0.11	0.08	0.10	0.08	0.09
	Nonpoor, Nonfishing	-0.02	-0.03	-0.02	-0.02	-0.02	-0.02
Population	Poor, Fishing	0.00	0.00	0.00	0.01	0.01	0.02
	Nonpoor, Fishing	0.00	0.00	0.00	0.00	0.00	0.00
	Poor, Nonfishing	0.00	0.00	0.00	0.01	0.01	0.03
	Nonpoor, Nonfishing	0.00	0.00	0.00	0.00	0.00	-0.01

Table S26: Percentage Changes (Relative to Baseline) in Outcomes for the FMP Intervention Under Alternative Annual Elasticities of Population. Fishing sector output prices are *exogenous*.

Outcome		Main text		$\eta = 0.025$		$\eta = 0.05$	
		Year 5	Year 10	Year 5	Year 10	Year 5	Year 10
$\mathcal{R}_0$		5.76	6.63	5.75	7.07	5.76	7.50
Fish Stock		41.16	42.57	41.16	42.57	41.16	42.57
Aggregate Output per Capita		1.75	2.34	1.76	1.82	1.75	1.33
Fish Harvest		10.00	12.63	10.01	12.63	10.01	12.63
Fishing Labour Time		23.39	27.21	23.39	27.23	23.38	27.26
Household Income per Capita	Poor, Fishing	16.68	19.20	16.70	17.54	16.69	15.98
	Nonpoor, Fishing	7.69	8.86	7.68	7.88	7.64	6.97
	Poor, Nonfishing	1.88	2.19	1.88	1.86	1.87	1.55
	Nonpoor, Nonfishing	0.75	0.88	0.75	0.70	0.74	0.54
Population	Poor, Fishing	0.00	0.00	-0.03	2.03	-0.01	4.00
	Nonpoor, Fishing	0.00	0.00	0.02	0.99	0.06	1.92
	Poor, Nonfishing	0.00	0.00	0.00	0.24	0.01	0.46
	Nonpoor, Nonfishing	0.00	0.00	0.00	0.10	0.01	0.18

Table S27: Percentage Changes (Relative to Baseline) in Outcomes for the TFP Intervention Under Alternative Annual Elasticities of Population. Fishing sector output prices are *exogenous*.

Outcome		Main text		$\eta = 0.025$		$\eta = 0.05$	
		Year 5	Year 10	Year 5	Year 10	Year 5	Year 10
$\mathcal{R}_0$		-2.41	-5.00	-2.31	-4.56	-2.22	-4.16
Fish Stock		0.11	0.26	0.11	0.26	0.11	0.25
Aggregate Output per Capita		4.44	9.30	4.27	8.54	4.12	7.87
Fish Harvest		0.00	0.07	0.01	0.07	0.01	0.07
Fishing Labour Time		-0.27	-0.51	-0.27	-0.48	-0.26	-0.46
Household Income per Capita	Poor, Fishing	3.71	7.83	3.54	7.05	3.38	6.36
	Nonpoor, Fishing	1.91	4.05	1.81	3.59	1.71	3.19
	Poor, Nonfishing	4.06	8.49	3.88	7.67	3.71	6.95
	Nonpoor, Nonfishing	2.60	5.45	2.47	4.85	2.34	4.33
Population	Poor, Fishing	0.00	0.00	0.17	0.78	0.34	1.48
	Nonpoor, Fishing	0.00	0.00	0.09	0.40	0.17	0.75
	Poor, Nonfishing	0.00	0.00	0.19	0.85	0.38	1.62
	Nonpoor, Nonfishing	0.00	0.00	0.12	0.55	0.24	1.02

Table S28: Percentage Changes (Relative to Baseline) in Outcomes for the MDA Intervention Under Alternative Annual Elasticities of Population. Fishing sector output prices are *endogenous*.

Outcome		Main text		$\eta = 0.025$		$\eta = 0.05$	
		Year 5	Year 10	Year 5	Year 10	Year 5	Year 10
$\mathcal{R}_0$		-44.55	-47.64	-44.55	-47.63	-44.55	-47.63
Fish Stock		-0.03	-0.05	-0.03	-0.05	-0.03	-0.05
Aggregate Output per Capita		0.02	0.03	0.02	0.02	0.02	0.01
Fish Harvest		0.00	-0.02	0.00	-0.02	0.00	-0.02
Fishing Labour Time		-4.23	-4.93	-4.23	-4.93	-4.23	-4.93
Household Income per Capita	Poor, Fishing	0.08	0.09	0.08	0.08	0.08	0.07
	Nonpoor, Fishing	0.00	-0.01	0.00	-0.01	0.00	-0.01
	Poor, Nonfishing	0.08	0.11	0.08	0.10	0.08	0.09
	Nonpoor, Nonfishing	-0.02	-0.03	-0.02	-0.02	-0.02	-0.02
Population	Poor, Fishing	0.00	0.00	0.00	0.01	0.01	0.03
	Nonpoor, Fishing	0.00	0.00	0.00	0.00	0.00	0.00
	Poor, Nonfishing	0.00	0.00	0.00	0.01	0.01	0.03
	Nonpoor, Nonfishing	0.00	0.00	0.00	0.00	0.00	-0.01

Table S29: Percentage Changes (Relative to Baseline) in Outcomes for the FMP Intervention Under Alternative Annual Elasticities of Population. Fishing sector output prices are *endogenous*.

Outcome		Main text		$\eta = 0.025$		$\eta = 0.05$	
		Year 5	Year 10	Year 5	Year 10	Year 5	Year 10
$\mathcal{R}_0$		3.74	4.55	3.80	4.88	3.86	5.19
Fish Stock		41.79	48.51	41.79	48.50	41.79	48.49
Aggregate Output per Capita		0.15	0.69	0.07	0.25	0.00	-0.14
Fish Harvest		6.08	12.45	6.08	12.46	6.08	12.46
Fishing Labour Time		13.73	17.47	13.74	17.51	13.74	17.54
Household Income per Capita	Poor, Fishing	10.09	13.10	9.81	11.79	9.54	10.61
	Nonpoor, Fishing	5.53	8.01	5.45	7.15	5.36	6.37
	Poor, Nonfishing	1.27	1.87	1.23	1.58	1.20	1.32
	Nonpoor, Nonfishing	0.67	1.10	0.66	0.94	0.66	0.80
Population	Poor, Fishing	0.00	0.00	0.39	1.72	0.78	3.32
	Nonpoor, Fishing	0.00	0.00	0.08	0.88	0.17	1.69
	Poor, Nonfishing	0.00	0.00	0.02	0.20	0.04	0.38
	Nonpoor, Nonfishing	0.00	0.00	-0.01	0.10	-0.02	0.18

Table S30: Percentage Changes (Relative to Baseline) in Outcomes for the TFP Intervention Under Alternative Annual Elasticities of Population. Fishing sector output prices are *endogenous*.

Outcome		Main text		$\eta = 0.025$		$\eta = 0.05$	
		Year 5	Year 10	Year 5	Year 10	Year 5	Year 10
$\mathcal{R}_0$		-2.11	-4.56	-2.00	-4.08	-1.90	-3.65
Fish Stock		-1.19	-3.23	-1.19	-3.24	-1.19	-3.25
Aggregate Output per Capita		4.83	10.03	4.65	9.22	4.48	8.50
Fish Harvest		0.05	-0.83	0.05	-0.83	0.05	-0.83
Fishing Labour Time		1.68	2.73	1.68	2.76	1.69	2.79
Household Income per Capita	Poor, Fishing	4.98	9.81	4.76	8.84	4.55	8.00
	Nonpoor, Fishing	2.09	4.00	1.97	3.51	1.86	3.09
	Poor, Nonfishing	4.13	8.53	3.94	7.68	3.76	6.94
	Nonpoor, Nonfishing	2.55	5.30	2.42	4.70	2.29	4.18
Population	Poor, Fishing	0.00	0.00	0.25	1.03	0.48	1.95
	Nonpoor, Fishing	0.00	0.00	0.11	0.43	0.20	0.80
	Poor, Nonfishing	0.00	0.00	0.20	0.86	0.39	1.64
	Nonpoor, Nonfishing	0.00	0.00	0.12	0.53	0.24	1.00

A9 Environmental Impact of Oil Palm

Oil palm cultivation has drawn widespread criticism for its environmental impacts, particularly in Southeast Asia, where expansion has contributed to deforestation, biodiversity loss, and greenhouse gas emissions [26, 27]. At the same time, oil palm generally produces more oil per hectare than other oil crops, supplying about 40% of global vegetable oil demand from less than 5.5% of the global oil crop area [28]. Oil palm produces about 81 million tonnes of oil from about 19 million hectares, whereas soybean and rapeseed together produce 84 million tonnes from over 163 million hectares [29]. Clearing rainforest or peatland for palm biodiesel creates a carbon debt that takes decades to centuries to repay, whereas biofuel crops grown on degraded or abandoned agricultural land incur little or no carbon debt [30]. Oil palm planted on degraded or already-cleared land may therefore represent a lower-emissions option for meeting rising demand for vegetable oils.

Environmental impacts vary widely across oilseed crops and depend heavily on local production systems. In Brazil, soybean expansion has been a major driver of deforestation in the Amazon and Cerrado biomes [31]. A life cycle assessment of five major vegetable oils finds that palm and soybean oil require the least land per tonne of oil and sunflower and peanut oil the most, while sunflower oil has the smallest water impact [32]. In contrast, Uganda's oil palm sector has developed on a more limited scale, with about 10,800 hectares planted in Kalangala District by 2016 [33], compared with about 19 million hectares worldwide [29], and under different ecological conditions. These differences suggest that the environmental footprint of oil palm in Uganda may be more modest or at least shaped by context-specific factors. Comparative assessments are therefore needed to evaluate oil palm's impacts alongside other oilseed crops and in light of Uganda's broader socio-ecological setting.

Incorporating Environmental Impacts into the Coupled Model

Integrating the environmental consequences of oil palm cultivation into the coupled model raises both conceptual and practical questions. One possible extension would be to make a parameter in the Eco or Epi components endogenous to oil palm output.

In the Eco component, for example, oil palm production could influence the lake's carrying capacity for fish (K_{lake}) or the intrinsic growth rate of the fish stock (r_{fstock}). This would reflect the possibility that nutrient runoff from oil palm could degrade water quality and, in turn, reduce the lake's ability to support fish, although studies of plantations in Indonesia generally report low nutrient losses and rarely assess effects at the watershed scale [34]. One way to represent this would be to express either K_{lake} or r_{fstock} as decreasing functions of oil palm output.

In the Epi component, a similar approach could be applied to parameters such as the household-specific exposure-risk parameter (ϵ_h). This could reflect more indirect pathways, such as the effect of water quality degradation on snail habitat suitability [13] or changes in human water contact. However, these links are harder to justify empirically and add conceptual complexity.

Implementing either version exceeds the scope of the current study. In both cases, meaningful progress would require data that are not currently available. Specifically, we would need empirical estimates of how changes in oil palm production translate into changes in nutrient loads, and how those, in turn, affect fish ecology or disease transmission. These relationships are likely nonlinear and difficult to quantify, especially in systems like Lake Victoria, where ecological baselines are shifting and long-term monitoring is limited [35]. Including these relationships would also increase the complexity of our coupled model in ways that could obscure the core mechanisms of interest. For these reasons, we do not incorporate these environmental relationships explicitly in the present analysis, but instead highlight them as an important area for future work.

A10 Figure: Lifecycle of the *Schistosoma* parasite

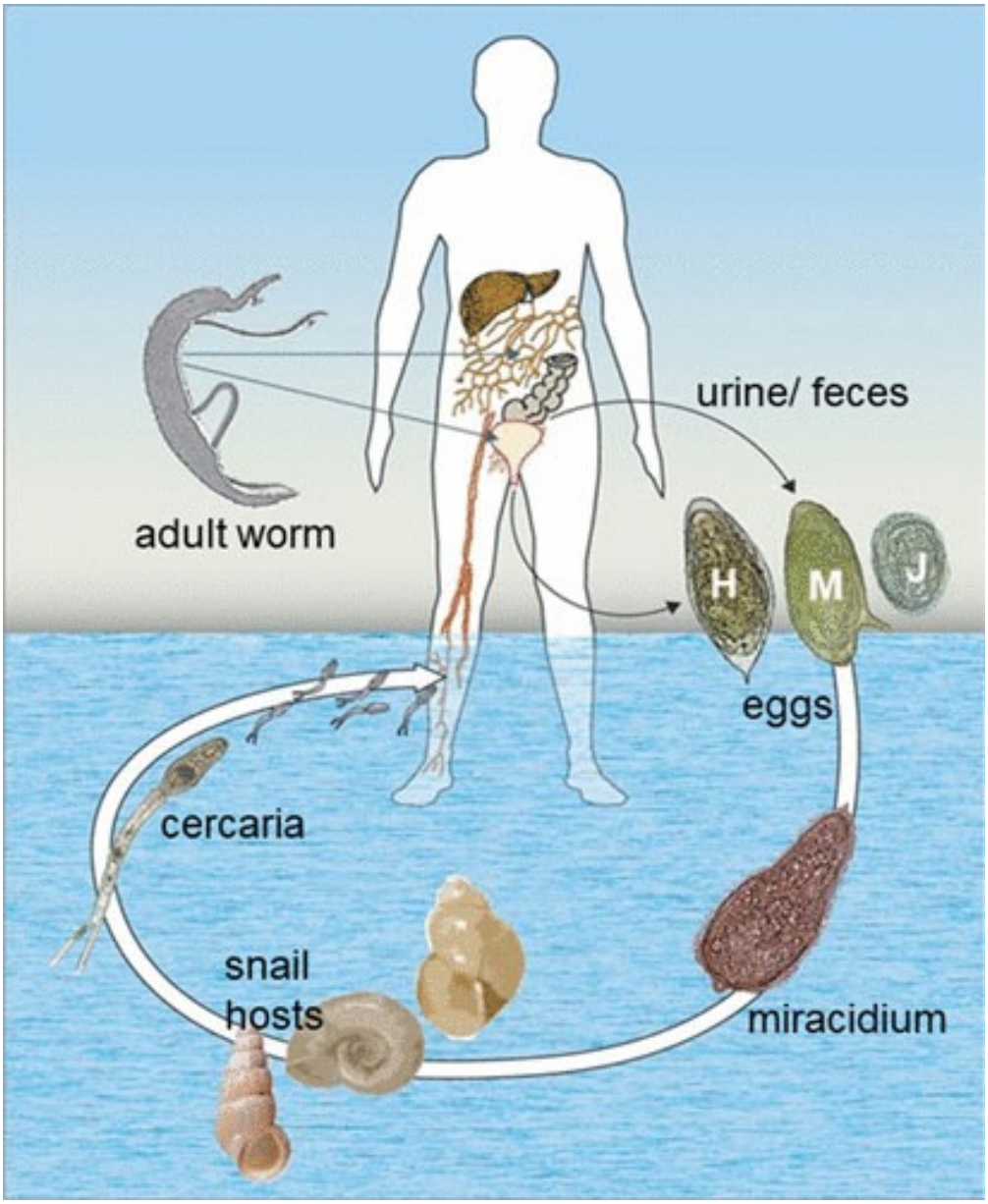


Figure S14: Lifecycle of the *Schistosoma* parasite. Adapted from Blanton, 2019 [36].

Bibliography

[1] UBOS, U. B. o. S. The national population and housing census 2014 – area specific profile series. Tech. Rep., Uganda Bureau of Statistics (2017).

[2] Nzira, J. E., Parlasca, M. C. & Qaim, M. An inquiry into labor conditions across key rural sectors in africa. *PLOS One* **20**, e0338694 (2025). URL http://dx.doi.org/10.1371/journal.pone.0338694.

[3] Mari, L. *et al.* Heterogeneity in schistosomiasis transmission dynamics. *Journal of Theoretical Biology* **432**, 87–99 (2017). URL https://linkinghub.elsevier.com/retrieve/pii/S0022519317303818.

[4] Garchitorena, A. *et al.* Disease ecology, health and the environment: A framework to account for ecological and socio-economic drivers in the control of neglected tropical diseases. *Philosophical Transactions of the Royal Society of London B: Biological Sciences* **372** (2017). URL http://rstb.royalsocietypublishing.org/content/372/1722/20160128. http://rstb.royalsocietypublishing.org/content/372/1722/20160128.full.pdf.

[5] Woolhouse, M. E. J., Etard, J.-F., Dietz, K., Ndhlovu, P. D. & Chandiwana, S. K. Heterogeneities in schistosome transmission dynamics and control. *Parasitology* **117**, 475–482 (1998).

[6] World Bank. Gdp, ppp (2017) (2022). URL https://data.worldbank.org/indicator/NY.GDP.MKTP.CD. Data retrieved from World Development Indicators.

[7] Gilliland, T. E., Sanchirico, J. N. & Taylor, J. E. A bioeconomic local general equilibrium assessment of distributional consequences of small-scale fisheries reform in developing countries. *Marine Resource Economics* **37**, 111–134 (2022). URL https://doi.org/10.1086/718379.

[8] Kateregga, E. & Sterner, T. Lake victoria fish stocks and the effects of water hyacinth. *The Journal of Environment & Development* **18**, 62–78 (2009). URL https://doi.org/10.1177/1070496508329467.

[9] Taylor, J. E., Whitney, E. & Zhu, H. Local-economy impacts of cash crop promotion. *Food and Agricultural Organization, United Nations* **40 IFAD Research Series**, 1–34 (2019). URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3465509.

[10] Bank, W. The uganda poverty assessment report 2016 - world bank (2016). URL https://pubdocs.worldbank.org/en/381951474255092375/pdf/Uganda-Poverty-Assessment-Report-2016.pdf.

[11] Taylor, J. E. & Filipski, M. J. (eds.) *Beyond Experiments in Development Economics: Local Economy-Wide Impact Evaluation* (Oxford University Press, Oxford, 2014).

[12] Sokolow, S. H. *et al.* Reduced transmission of human schistosomiasis after restoration of a native river prawn that preys on the snail intermediate host. *Proceedings of the National Academy of Sciences* **112**, 9650–9655 (2015). URL https://www.pnas.org/content/112/31/9650.

[13] Halstead, N. T. *et al.* Agrochemicals increase risk of human schistosomiasis by supporting higher densities of intermediate hosts. *Nature Communications* **9** (2018).

[14] Moira, A. P. D. *et al.* Microgeographical and tribal variations in water contact and schistosoma mansoni exposure within a ugandan fishing community. *Tropical Medicine & International Health* **12**, 724–735 (2007).

[15] Diekmann, O., Heesterbeek, J. A. P. & Roberts, M. G. The construction of next-generation matrices for compartmental epidemic models. *Journal of The Royal Society Interface* **7**, 873–885 (2009).

[16] Castonguay, F. M., Sokolow, S. H., Leo, G. A. D. & Sanchirico, J. N. Cost-effectiveness of combining drug and environmental treatments for environmentally transmitted diseases. *Proceedings of the Royal Society B: Biological Sciences* **287**, 20200966 (2020).

[17] King, C. H. *et al.* Impact of different mass drug administration strategies for gaining and sustaining control of schistosoma mansoni and schistosoma haematobium infection in africa. *The American Journal of Tropical Medicine and Hygiene* **103**, 14–23 (2020). URL https://doi.org/10.4269/ajtmh.19-0829.

[18] Fenwick, A. & Figenschou, B. H. The effect of schistosoma mansoni infection of the productivity of cane cutters on a sugar estate in tanzania. *Bull World Health Organ* **47**, 567–572 (1972). URL https://www.ncbi.nlm.nih.gov/pubmed/4540675.

[19] Barbosa, F. S. & da Costa, D. P. P. Incapacitating Effects of Schistosomiasis Mansoni on the Productivity of Sugar-cane Cutters in Northeastern Brazil. *American Journal of Epidemiology* **114**, 102–111 (1981). URL https://doi.org/10.1093/oxfordjournals.aje.a113156. http://oup.prod.sis.lan/aje/article-pdf/114/1/102/226727/114-1-102.pdf.

[20] Colley, D. G., Bustinduy, A. L., Secor, W. E. & King, C. H. Human schistosomiasis. *Lancet* **383**, 2253–2264 (2014). URL https://www.ncbi.nlm.nih.gov/pubmed/24698483.

[21] King, C. H., Sturrock, R. F., Kariuki, H. C. & Hamburger, J. Transmission control for schistosomiasis – why it matters now. *Trends in Parasitology* **22**, 575–582 (2006). URL http://dx.doi.org/10.1016/j.pt.2006.09.006.

[22] Gurarie, D., King, C. H., Yoon, N. & Li, E. Refined stratified-worm-burden models that incorporate specific biological features of human and snail hosts provide better estimates of schistosoma diagnosis, transmission, and control. *Parasites & Vectors* **9** (2016).

[23] Todaro, M. P. Internal migration in developing countries: A survey. In Easterlin, R. A. (ed.) *Population and Economic Change in Developing Countries*, 361–402 (University of Chicago Press, Chicago, 1980). National Bureau of Economic Research volume. ISBN 0-226-18027-1.

[24] Hirvonen, K. Temperature changes, household consumption, and internal migration: Evidence from tanzania. *American Journal of Agricultural Economics* **98**, 1230–1249 (2016).

[25] Bazzi, S. Wealth heterogeneity and the income elasticity of migration. *American Economic Journal: Applied Economics* **9**, 219–255 (2017).

[26] Koh, L. P. & Wilcove, D. S. Is oil palm agriculture really destroying tropical biodiversity? *Conservation Letters* **1**, 60–64 (2008). URL http://dx.doi.org/10.1111/j.1755-263X.2008.00011.x.

[27] Carlson, K. M. *et al.* Committed carbon emissions, deforestation, and community land conversion from oil palm plantation expansion in west kalimantan, indonesia. *Proceedings of the National Academy of Sciences* **109**, 7559–7564 (2012). URL http://dx.doi.org/10.1073/pnas.1200452109.

[28] Meijaard, E. *et al.* The environmental impacts of palm oil in context. *Nature Plants* **6**, 1418–1426 (2020).

[29] Murphy, D. J., Goggin, K. & Paterson, R. R. M. Oil palm in the 2020s and beyond: challenges and solutions. *CABI Agriculture and Bioscience* **2**, 39 (2021).

[30] Fargione, J., Hill, J., Tilman, D., Polasky, S. & Hawthorne, P. Land clearing and the biofuel carbon debt. *Science* **319**, 1235–1238 (2008).

[31] Gibbs, H. K. *et al.* Brazil's Soy Moratorium. *Science* **347**, 377–378 (2015).

[32] Schmidt, J. H. Life cycle assessment of five vegetable oils. *Journal of Cleaner Production* **87**, 130–138 (2015).

[33] International Fund for Agricultural Development. Republic of Uganda Vegetable Oil Development Project, Phase 2 (VODP2): Supervision report, main report and appendices. Tech. Rep. 4096-UG,

International Fund for Agricultural Development, Rome (2016). Mission dates 9 to 20 May 2016, document dated 2 June 2016. No DOI.

[34] Comte, I., Colin, F., Whalen, J. K., Grünberger, O. & Caliman, J.-P. Agricultural practices in oil palm plantations and their impact on hydrological changes, nutrient fluxes and water quality in Indonesia: a review. In Sparks, D. L. (ed.) *Advances in Agronomy*, vol. 116, 71–124 (Academic Press, 2012).

[35] Verschuren, D. *et al.* History and timing of human impact on lake victoria, east africa. *Proceedings of the Royal Society of London. Series B: Biological Sciences* **269**, 289–294 (2002). URL http://dx.doi.org/10.1098/rspb.2001.1850.

[36] Blanton, R. E. Population structure and dynamics of helminthic infection: Schistosomiasis. *Microbiology Spectrum* **7**, 7.4.16 (2019). URL https://journals.asm.org/doi/abs/10.1128/microbiolspec.AME-0009-2019. https://journals.asm.org/doi/pdf/10.1128/microbiolspec.AME-0009-2019.